

Deep research report

Data, open-source projects, current technology and integration paths for the AquaSync dam–river digital twin

Produced by a 13-agent research sweep across six domains.

Every URL curl-verified; every claim checked by a second, adversarial agent.

148 sources confirmed after verification	0 claims rejected as unverifiable	14 resources downloaded into research/	561.0 MB acquired and indexed locally
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Why every link in this document was verified by curl.

The planning material this project grew out of was full of confident, wrong references: one Amazon ASIN appeared as five different products, and a dataset was described as containing 2018 flood data it does not contain. Both errors survived because nobody checked.

So each research agent was required to run `curl -o /dev/null -w "%{http_code}"` against every URL and record the real status, and a second agent then re-checked the list adversarially and rejected what it could not confirm. Every claimed source survived that pass.

Generated 26 August 2026 · Regenerate with `scripts/build_research_report.py` · Local index at `research/index/README.md`

5 · Historical Kerala / India Reservoir And Flood Data, Especially Pre 2020 (Idukki, Idamalayar, Periyar Basin)

Confirmed sources (30)

Source	Kind	What it provides	Relevance	HTTP
National Water Data Portal (NWDP) CKAN API live; Kera · Government Open Data License - India (portal-level) https://nwdp.nwic.gov.in/api/3/action/package_list	api	Open CKAN 2.x API over 555 datasets from CWC, CGWB and every state water department. No registration, no API key, no rate limit hit in ~40 calls. Endpoints used and verified:... caveat: Dataset titles routinely lie about temporal coverage (see the Kerala SW discharge entry). Always download and histogram the date column...	high - this is the single highest-value new door for AquaSync; it is how you reach every...	200
CWC Kerala daily river discharge 2001-2025 (NWDP resource) contains 2 · GODL-India https://nwdp.nwic.gov.in/dataset/08fa3fd0-7861-471d-a295-...	dataset	39,425,558 bytes, 217,538 rows, 40 Kerala CWC gauging stations. Periyar rows = 30,833. NEELEESWARAM 8,478 records 2001-2025 (284 in 2018); VANDIPERIYAR 14,837 (568 in 2018, twice-daily 08:00 and... caveat: VERIFIED HOLE: Neeleeswaram August 2018 is missing days 5,12,16,17,18,19,20,21,22,24,25,26,27 - the flood peak (16-22 Aug) is entirely...	high - genuine observed Periyar discharge covering 2018 and the whole pre-2020 period;...	200
CWC Kerala daily river discharge 1950-2000 (NWDP resource) static his · GODL-India https://nwdp.nwic.gov.in/dataset/08fa3fd0-7861-471d-a295-...	dataset	23,703,774 bytes, 134,069 rows. Periyar rows = 19,588. NEELEESWARAM 10,884 records starting 1971 (near-complete 365/yr 1972-1977, then 730/yr from 1978 as two readings per day); ARANGALI 8,276;... caveat: Despite the '1950' in the filename, the earliest Periyar record is 1971. Deduplicate against the 2001-2025 file - year 2000 appears in both.	high - gives a 55-year continuous Periyar discharge record when joined to the 2001-2025...	200
Kerala SW manual daily rainfall 1991-2020 (NWDP resource) companion · GODL-India https://nwdp.nwic.gov.in/dataset/c379c7dd-755c-4a0d-9de1-...	dataset	24,200,224 bytes, 151,429 rows. Actual year span is 2000-2020 (not 1991). 2018 has 8,435 station-days. IDUKKI district 7,462 rows, ERNAKULAM 366. Long-record catchment stations: BOYCE ESTATE... caveat: Titled 1991-2020, actually starts 2000. Only 2 of the 7 Idukki/Ernakulam stations have multi-year records - Neriya Mangalam and...	high - real gauge rainfall inside the Idukki catchment spanning 2018, to force the...	200
Kerala SW manual daily river discharge (NWDP) - NEGATIVE, do not trust the label listed as · GODL-India https://nwdp.nwic.gov.in/dataset/river-discharge-manual-d...	dataset	Advertised as 'River Discharge Kerala_SW Kerala (2001 - 2025) Manual Daily'. Actually 765,060 bytes, 4,904 rows, ALL from calendar year 2011, 23 stations, ZERO Periyar rows. caveat: Worked example of the NWDP metadata problem. Stations are also mislabelled with rivers from other states (Narmada, Mahi, Luni, Kim, Banas,...	low - recorded so nobody else wastes a day on the promising title; it is the...	200

Source	Kind	What it provides	Relevance	HTTP
RAT 3.0 Kerala live dashboard (CWRDM + UW SASWE + NASA) data ends · not stated on the portal; RAT engine itself is GPL-3.0 http://210.212.226.241:8080/RAT_Kerala_CWRDM/	tool	Satellite + VIC reservoir monitoring for 19 Kerala reservoirs including Idukki (9.845,76.968) and Idamalayaar (10.222,76.706). Products per dam: inflow, outflow, area-elevation curve, TMS-OS surface... caveat: Bare IP over plain HTTP on port 8080 - no TLS, so it will fail any HTTPS-only fetcher (this is why WebFetch errored with...	high - the only downloadable daily Idukki/Idamalayar inflow series that spans 2018, plus...	200
RAT Kerala - Idukki daily inflow CSV 2025-02-22 · unstated http://210.212.226.241:8080/RAT_Kerala_CWRDM/assets/RAT_r...	dataset	100,264 bytes, 3,410 daily rows, columns date,streamflow (m3/s), 2015-10-24 to 2025-02-22, unbroken. August 2018 complete (31/31 days). Idamalayar twin at .../inflow/Idamalayaar.csv, same span. caveat: VERIFIED MISMATCH at the event that matters: RAT Aug-2018 peak 1104.6 m3/s on the 18th vs CWC-observed 2532 cumec at 22:00 on the 15th...	high - directly fills 'no pre-2020 Idukki data' as a MODELLED series; usable as a prior...	200
RAT Kerala - Idukki area-elevation curve (AEC) static · unstated http://210.212.226.241:8080/RAT_Kerala_CWRDM/assets/RAT_r...	dataset	626 rows at 1 m elevation steps, 551 m to 1176 m, columns elevation,area(km2). Idukki: 50.015 km2 @730 m, 51.098 @731, 52.125 @732, 53.201 @733. Idamalayar twin has 588 rows, 62-649 m: 30.159 @167,... caveat: DEM-derived, and it disagrees with the literature at FRL: it gives 52.1 km2 at 732 m whereas Suresh et al. 2024 Table 1 lists Idukki as 60...	high - an independent hypsometric curve to cross-validate AquaSync's $r^2=0.9957...$	200
RAT Kerala - Idukki TMS-OS surface area and storage change 2025-02-21 · unstated http://210.212.226.241:8080/RAT_Kerala_CWRDM/assets/RAT_r...	dataset	657 rows 2017-07-29 to 2025-02-21 (85 observations in 2018), columns date,area (km2). Paired storage change at /dels/Idukki.csv, columns date,dS (m3). Optical (MNDWI/NDWI) fused with SAR. caveat: THE AUGUST 2018 VALUES ARE NOT REAL OBSERVATIONS. Area reads 46.2 km2 on 12 Aug then an identical 44.373 km2 on 17, 22 AND 27 Aug - a...	medium - real satellite observation of Idukki through 2018, but see caveat; more...	200
CWC Reservoir Level & Storage Bulletin archive published · Government of India, free to reuse https://cwc.gov.in/en/reservoir-level-storage-bulletin	doc	Weekly (Thursday) PDF bulletins covering 161 major Indian reservoirs, Idukki and Idamalayar among them. Per-reservoir fields: FRL (m), live capacity (BCM), bulletin date, current level (m), current... caveat: Weekly only - far too coarse to constrain a flood-routing model. Filenames are wildly inconsistent (DD.MM.YYYYcwcbull.pdf, ..._1.pdf,...	high - the only official numeric Idukki level+storage series I could actually retrieve...	200

Source	Kind	What it provides	Relevance	HTTP
CWC weekly bulletin, 02 August 2018 (worked example) static 201 · Government of India https://cwc.gov.in/sites/default/files/02.08.2018cwcbull_...	doc	539,052 bytes. Verified extracted row: 'IDUKKI 732.43 730.25 1.460 1.338 01-08-2018 92 23 41 - 780' = FRL 732.43 m, level 730.25 m, live cap 1.460 BCM, live storage 1.338 BCM, 92% full, 23% same... caveat: Idukki appears twice per bulletin in two different table layouts (state table and the 161-reservoir national table) - anchor your regex or...	high - concrete proof the 2018 Idukki level/storage numbers are retrievable, and the FRL...	200
CWC 'Kerala Floods of August 2018' study report (via KSDMA) static (pu) · Government of India https://sdma.kerala.gov.in/wp-content/uploads/2020/08/CWC...	doc	48 pages, 6,003,314 bytes. THE primary source (it is reference 4 of the Current Science paper). Verified quantitative content: Idukki peak inflow 2532 cumec at 22:00 on 15-08-2018; release 1614... caveat: Fig.4 (Idukki inflow/outflow/level) and Fig.5 (Idamalayar) are CHARTS - there is no data table and no annexure. BUT the x-axis tick labels...	high - the authoritative 2018 numbers, and the correct calibration/validation targets...	200
Sudheer et al. 2019, 'Role of dams on the floods of August 2018 in Periyar River Basin, Kerala', Current Science 116(5):780-794 published · Current Science, free to read https://www.currentscience.ac.in/Volumes/116/05/0780.pdf	paper	Free full PDF, 15 pages, 4,413,605 bytes. HEC-HMS reconstruction of the whole Periyar basin. Verified numbers: simulated Idukki peak inflow 2763 m3/s at 20:00 on 15-08-2018 vs measured 2532 m3/s;... caveat: NO DATA TABLE of any hydrograph. The observed 6-hourly Idukki inflow/release appears only inside Fig.6a and is cited to the CWC report....	high - the closest published analogue to what AquaSync's optimiser does, and its 16-21%...	200
Suresh et al. 2024, RSE - free full manuscript on NASA NTRS 2024 · NASA-funded manuscript, publicly posted (Grant 80NSSC22K0918) https://ntrs.nasa.gov/api/citations/20240007715/downloads...	paper	56 pages, 3,714,034 bytes - the accepted manuscript of the paywalled Remote Sensing of Environment paper (DOI 10.1016/j.rse.2024.114149), free. Contains the full 19-reservoir table with coordinates... caveat: Accepted manuscript, so figure numbering may differ slightly from the published version. The observed KSEB series it validates against is...	high - gives the exact, citable error bars for the RAT inflow series AquaSync would...	200
Minocha et al. 2024, 'Reservoir Assessment Tool version 3.0', Geoscientific Model Development 17:3137 2024-04-23 · CC-BY 4.0 https://gmd.copernicus.org/articles/17/3137/2024/	paper	Open-access description of the RAT 3.0 software platform: VIC hydrological model + MetSim + routing, IMERG forcing, TMS-OS surface area, GEE backend. Crossref-verified: published 2024-04-23, 6... caveat: Software paper, contains no Kerala data itself.	medium - the method paper you need if AquaSync ever runs its own RAT instance over the...	200
UW-SASWE/RAT (Reservoir Assessment Tool source) 26★ · last push · GPL-3.0 https://github.com/UW-SASWE/RAT	repo	Python implementation of RAT 3.0 - VIC + MetSim + routing + Google Earth Engine surface-area extraction, producing inflow, outflow, storage change, evaporation and AEC per reservoir. caveat: Only 26 stars and GPL-3.0, which is a licence constraint if AquaSync ships closed. Heavy stack: needs a Google Earth Engine service...	medium - running it yourself over the Periyar with IMD rather than IMERG forcing, and...	200

Source	Kind	What it provides	Relevance	HTTP
IMD 0.25-degree daily gridded rainfall - direct POST endpoint extended a · IMD open data (cite Pai et al. 2014) https://www.imdpune.gov.in/cmpg/Griddata/RF25.php	api	The form action behind the IMD download page. POST body RF25= returns one NetCDF per year. VERIFIED LIVE: RF25=2018 gave HTTP 200, 25,431,832 bytes, CDF-1 classic NetCDF, variables... caveat: 0.25 deg is ~28 km - the Idukki catchment (~1240 km2 incl. free catchment) is only a handful of cells, so orographic gradients across the...	high - the long-record rainfall forcing AquaSync's SCS-CN model needs for any pre-2020...	200
IMD gridded rainfall NetCDF landing page lists year · IMD open data https://www.imdpune.gov.in/cmpg/Griddata/Rainfall_25_NetC...	doc	Human-facing year-picker for the 0.25 deg 1901-2024 daily rainfall product; documents the 135x129 grid layout and mm units. Contact for the group: cmagpune@gmail.com. caveat: Page itself exposes no download URLs to a scraper - you must know the RF25.php action. Sibling products exist for 1.0 deg rainfall and for...	medium - the citable landing page; the actual bytes come from RF25.php above	200
imdlb (Python client for IMD gridded data) 43★ · PyPI relea · MIT https://pypi.org/project/imdlb/	tool	Downloads and parses IMD gridded rainfall and temperature into xarray; handles the binary GRD variants as well as NetCDF. Version 0.1.21, MIT licensed. caveat: Small project (43 stars). Its download routines break whenever IMD changes the form endpoint - verify against the raw curl before trusting...	medium - saves writing the NetCDF/GRD parsing, though the raw curl above is a one-liner...	200
GUARDIAN: Extending Sub-Daily River Discharge data over India (Scientific Data, 2024) data depos · CC-BY (Scientific Data) https://doi.org/10.1038/s41597-024-03923-8	paper	Hourly (monsoon Jun-Sep) and sub-daily (non-monsoon) discharge for 210 CWC stations, 2020 to present, built by scraping water-surface elevation off the CWC flood portal and applying station rating... caveat: STARTS 2020 - it does nothing for the 2018 gap. Discharge is rating-curve-derived, not measured, so it inherits rating error (Neeleeswaram...)	high - the practical answer to AquaSync gap #2 (Muskingum K and x never...	200
GUARDIAN data on figshare (data.rar + rating curves) 2024-10-19 · figshare API reports no licence field set - clarify before redistributing https://doi.org/10.6084/m9.figshare.27004282	dataset	Two files, confirmed via the figshare API (article 27004282, published 2024-10-19): data.rar 577.0 MB (extended WSE + extended discharge per station) and RC.xlsx 0.1 MB (rating curves). I downloaded... caveat: HTTP 202 is figshare's normal async-landing response, not an error - the ndownloader links return 200. Archive is .rar, needs unrar/7z....	high - the rating curves alone (0.1 MB) let AquaSync convert any CWC stage reading to...	200
GHI: Geospatial dataset for hydrologic analyses in India (ESSD 15:4389, 2023) 2023-10-05 · CC-BY 4.0 https://essd.copernicus.org/articles/15/4389/2023/	dataset	Quality-controlled CWC river gauges with catchment boundaries and hydrometeorological time series over 71 water years (1950-2020). Of 645 CWC stations screened: 213 Group-1 (reliable metadata +... caveat: The data itself is on Zenodo, which returned 403 'unusual traffic from your network' for EVERY Zenodo URL I tried from this machine. The...	high - delineated catchment polygons for the Periyar CWC gauges would let AquaSync...	200

Source	Kind	What it provides	Relevance	HTTP
CAMELS-IND (ESSD 17:461, 2025) 2025-02-05 · CC-BY 4.0 https://essd.copernicus.org/articles/17/461/2025/	dataset	472 Peninsular-India catchments (228 with observed streamflow for >30% of 1980-2020), 1 Jan 1980 - 31 Dec 2020 daily. 19 meteorological forcings (precip, tmax/tmin, LW/SW radiation, U/V wind, RH,... caveat: Zenodo returned 403 from this network for every attempt (doi.org redirect, /records/, and the /api/records/ JSON endpoint), so I could NOT...	medium-high - a ready large-sample training set if AquaSync attacks gap #3...	200
Mishra et al., 'The Kerala flood of 2018: combined impact of extreme rainfall and reservoir storage' (HESS Discussions preprint) 2018 prepr · CC-BY 4.0 (Copernicus preprint) https://hess.copernicus.org/preprints/hess-2018-480/	paper	Open preprint with full PDF. Establishes that 1-15 day August 2018 rainfall upstream of Idukki, Kakki and Periyar reservoirs had >500-year return periods, and that 6 of 7 major Kerala reservoirs... caveat: Preprint only - never completed HESS peer review, so cite with care. Its Idukki storage series stops on 8 August 2018, i.e. it stops one...	medium - useful for antecedent-condition framing and it independently confirms...	200
India-WRIS API contract, recovered from WRIS_India_data_extractor source 0★ · created 20 · none declared https://github.com/DVTanna369/WRIS_India_data_extractor	repo	Flask app that wraps the undocumented India-WRIS endpoints. Reading the source gives the complete, otherwise-undocumented contract: POST https://indiawris.gov.in/DataSet/DataSetList ; POST... caveat: 0 stars, one-week-old project, no licence, effectively unmaintained. AND the target host is dead from this network - see dead_ends. The...	medium - the endpoint spec is worth keeping even though the host is unreachable from...	200
amith-vp/Kerala-Dam-Water-Levels (already in AquaSync - refreshed metrics) 54★ · pushed 202 · NONE DECLARED (GitHub API returns license: null) https://github.com/amith-vp/Kerala-Dam-Water-Levels	repo	KSEB + Irrigation bulletin scrape, live and historical JSON, GitHub Actions updated. Confirmed via authenticated GitHub API today: 54 stars, last push 2026-08-26T08:05:16Z (same day), repo size... caveat: CONFIRMS THE PROJECT'S EXISTING FINDING: this repo cannot contain 2018 data - GitHub Actions only began populating it in Aug 2020....	high - already integrated; recorded here only to confirm it is still actively pushing...	200
GloLakes: water storage dynamics for 27,000 lakes, 1984-present (ESSD 16:201, 2024) 2024-01-11 · CC-BY 4.0 (paper) https://essd.copernicus.org/articles/16/201/2024/	dataset	Relative and absolute water storage time series from radar altimetry (Topex/Poseidon, Jason-1/2/3, Sentinel-3/6), ICESat-2 laser altimetry and Landsat/Sentinel-2 optical, 1984 to present... caveat: I could NOT confirm Idukki or Idamalayar are in the 27,000 - the NCI GeoNetwork landing page is a JS catalogue and I did not open the...	medium - a genuinely independent satellite storage series that spans 2018, worth...	200

Source	Kind	What it provides	Relevance	HTTP
Global Reservoir Surface Area Dataset (GRSAD) static; su · Texas Data Repository terms https://dataverse.tdl.org/dataset.xhtml?persistentId=doi:...	dataset	Monthly Landsat-derived surface area for 6,817 global reservoirs (GRanD-matched) with cloud/shadow/SLC-off contamination repaired. March 1984 - October 2015. caveat: Hard stop at 2015-10 makes it useless for the 2018 event. Monthly resolution is in any case far too coarse for flood routing. Its only...	low - ENDS OCTOBER 2015, so it cannot see 2018 at all; listed only to close the question	200
KSDMA dam water level page live · Government of Kerala https://sdma.kerala.gov.in/dam-water-level/	doc	Kerala State Disaster Management Authority's daily water-level page for major KSEB and Irrigation dams; also hosts the mirrored CWC 2018 flood report and a /floods_2018/ section (verified 200). caveat: Presents current levels only - no historical archive, no API, no bulk download. This is the same source the amith-vp scraper consumes, so...	medium - KSDMA is the institutional owner and the most likely formal counterparty for a...	200
CWRDM - Centre for Water Resources Development and Management, Kerala live; RAT- · Government of Kerala https://cwrwm.kerala.gov.in/reservoir-assessment-tool-ker...	doc	Kerala state R&D; institute under KSCSTE that co-developed RAT 3.0 Kerala with UW SASWE under NASA grant 80NSSC22K0918. Hosts the RAT-Kerala dashboard and its documentation. caveat: No data is published by CWRDM directly beyond the RAT dashboard CSVs - obtaining the underlying KSEB 2018 series requires a formal...	high - as an institutional route this is the best lead for AquaSync gaps #1 and #5:...	200

Verification notes

- This domain has NOT been through the adversarial verification pass. URLs carry the researching agent's own reported status only. Treat as provisional.
- dead end: indiawris.gov.in IS COMPLETELY UNREACHABLE from this network - not slow, dead. DNS resolves fine (164.100.85.36) but TCP connect never completes: 21 s to timeout, curl exit 28, HTTP code 000. Tested <https://indiawris.gov.in/>, <http://indiawris.gov.in/>, <https://www.indiawris.gov.in/>, <https://indiawris.gov.in/wris/>, <https://indiawris.gov.in/swagger-ui/index.html>,...
- dead end: NWDP HAS NO KERALA RESERVOIR LEVEL OR STORAGE DATA AT ALL. I enumerated all 555 packages. Of 70 'reservoir' packages, 62 are `reservoir_discharge_*_telemetry_hourly` for individual dams in MP, Rajasthan, Maharashtra, Telangana, UP, WB, Uttarakhand and Punjab - not one is in Kerala. The only reservoir level/storage packages are for Andhra Pradesh, BBMB Chandigarh, Uttarakhand, West...
- dead end: [data.gov.in](https://www.data.gov.in) is currently serving a SANDBOX banner and its API is broken for this dataset. The page <https://www.data.gov.in/resource/daily-data-reservoir-level-central-water-commission-cwc> returns 200 to curl but 403 to WebFetch, is a JS shell, and carries the literal text 'This is a sandbox environment created for testing and demonstration purposes only. The content, data, and...
- dead end: ZENODO IS 403 FROM THIS NETWORK - which blocks two otherwise-excellent datasets. <https://doi.org/10.5281/zenodo.14005378> (CAMELS-IND), <https://zenodo.org/records/14005378>, <https://zenodo.org/api/records/14005378> and <https://doi.org/10.5281/zenodo.7563599> (GHI) all return 403 with the body 'Access to this resource has been restricted due to unusual traffic from your network.' The papers...
- dead end: ffs.india-water.gov.in (CWC flood forecasting) has NO usable historical archive or API surface. Root returns 200 but it is an Angular SPA (`styles.23a61c3e13ab6949c945.css`, Kendo UI) with zero crawlable

sub-paths: /ffs, /reservoir, /reservoir-level, /ReservoirLevel, /flood-forecast, /api, /dashboard, /static and /home ALL return 404. It serves live forecasts only. The one thing it is...

- dead end: The two CWC weekly bulletins covering the 2018 flood peak DO NOT EXIST in the public archive. 45 of 52 Thursdays in 2018 are listed and downloadable, but 2018-08-09 and 2018-08-16 are absent (as are 03-29, 05-10, 09-27, 10-18, 12-27). I tried seven filename variants for each - 09.08.2018cwcbull.pdf, 09.08.2018cwcbull_1.pdf, 09.08.2018cwc.pdf, 09.08.2018_0.pdf, 09.08.2018cwcbull_0.pdf,...
- dead end: GRSAD (Global Reservoir Surface Area Dataset) ENDS OCTOBER 2015 and is monthly. It cannot see the 2018 flood, and monthly resolution is useless for routing regardless. Ruled out.
- dead end: GRDC (Global Runoff Data Centre) is not a route to Indian discharge. The portal is up (portal.grdc.bafg.de 200) but the literature is explicit that GRDC and RivDIS hold only legacy data for a small fraction of CWC stations, with no provenance and unknown gaps, and that CWC classifies transboundary-basin data. Two catalogue URLs I tried are also dead:...
- dead end: keralafloodwatch.in and dams.keralam.co are LIVE DASHBOARDS ONLY (both 200). keralafloodwatch shows current Idukki and Idamalayar storage percentages and gate status and names IMD/CWC/KSDMA/KSEB as sources, but exposes no API, no bulk download, and no confirmed pre-2020 archive. They re-present data AquaSync already ingests via amith-vp. No new information.
- dead end: MDPI and ScienceDirect both 403 this network, so several promising papers could not be read directly:
<https://www.mdpi.com/2073-4441/15/7/1371> (403),
<https://www.sciencedirect.com/science/article/abs/pii/S0034425724001603> (403),
<https://www.sciencedirect.com/science/article/pii/S2214581825000862> (403). These are BOT BLOCKS, not dead links - I confirmed 10.3390/w15071371 and...

6 · Open Source Reservoir Operation, Optimisation And Forecast Informed Operations (Firo)

Confirmed sources (29)

Source	Kind	What it provides	Relevance	HTTP
rtc-tools/rtc-tools (GitHub) 24★ · 2026-08-26 · LGPL-3.0 https://github.com/rtc-tools/rtc-tools	repo	The RTC-Tools framework: goal programming, ensemble/scenario-tree optimisation (ControlTreeMixIn), Modelica-based system modelling on CasADi. 16,410 lines of Python in src/. 15 shipped examples... caveat: 24 stars understates it (LF Energy project with Deltares commercial backing) but does mean a thin community for troubleshooting. I...	high - closest thing to a production-grade open reservoir MPC framework; its...	200
rtc-tools on PyPI 2.8.0 on 2 · LGPL-3.0 https://pypi.org/project/rtc-tools/	tool	Version 2.8.0 (2026-07-08), requires-python >=3.10, 88 releases. Declared deps: casadi!=3.6.6,<3.8,>=3.6.3, defusedxml>=0.7.0, numpy>=1.21.2, pymoca==0.9.*, rtc-tools-channel-flow>=1.2.0,... caveat: Pins casadi<3.8 while CasADi is at 3.8.0, and pymoca==0.9.* while pymoca is at 0.12.0. Install still clean. The readthedocs Getting...	high - I used this to MEASURE the real dependency burden rather than guessing at it	200
Robust-FIRO (Taylor & Herman, UC Davis) 1★ · 2025-03-18 · MIT https://github.com/wltaylor/Robust-FIRO	repo	MPCProblem.py - a 194-line cvxpy receding-horizon ensemble MPC for reservoir flood control. Also train_EFO.py (risk-threshold Ensemble Forecast Operations policy search), model.py (13 KB simple... caveat: Paper-support code, not a library: 1 star, no packaging, no tests, hard-coded to Oroville. `import mosek` at module level, and a...	high - THE most useful artefact in this dimension; the concrete answer to the...	200
SSJRB-FIRO-Publishing (Taylor & Herman) 1★ · 2024-05-20 · MIT https://github.com/wltaylor/SSJRB-FIRO-Publishing	repo	Multi-reservoir FIRO across the Sacramento-San Joaquin River Basin. Contains perfect_forecast.py (an explicit perfect-foresight baseline - exactly AquaSync's current assumption, isolated so the gap... caveat: 1 star, research code, California-specific data. Value is the experimental design and LOOCV validation pattern, not runnable-as-is software.	high - the perfect_forecast.py vs simulate_firo.py pairing is the exact experimental...	200
zpb4/syn-forecast_stochastic (Brodeur) 1★ · 2026-08-19 · MIT https://github.com/zpb4/syn-forecast_stochastic	repo	Synthetic hydrology and synthetic ensemble-forecast generation plus a FIRO model for Lake Oroville, supporting a manuscript on FIRO under climate extremes and evolving forecast skill. caveat: 1 star, 196 KB. Companion R repo zpb4/FIRO_synthetic-ensemble-forecasts (verified 200, 0 stars, no licence, 2024-01) implements the...	high - AquaSync has NO operational ensemble forecast feed for Kerala; this is the method...	200

Source	Kind	What it provides	Relevance	HTTP
Lake Mendocino FIRO Final Viability Assessment (Dec 2020) December 2 https://cw3e.ucsd.edu/FIRO_docs/LakeMendocino_FIRO_FVA.pdf	standard	The primary published FVA for the flagship deployed FIRO programme; basis for USACE approving FIRO-based operations at Coyote Valley Dam. Produced by Sonoma Water + CW3E/Scripps with federal, state... caveat: The commonly-cited eScholarship mirror (escholarship.org/uc/item/3b63q04n) returns 403 to curl - use this CW3E PDF. Contains no code....	high - the authoritative precedent for arguing forecast-informed operation to a...	200
Prado Dam FIRO Final Viability Assessment https://cw3e.ucsd.edu/FIRO_docs/Prado/FIRO_Prado_FVA.pdf	standard	The second completed FVA. Prado FIRO confirmed viable (2023), implementation ongoing including a water control manual update; reported benefit up to 23,000 acre-feet of additional recharge in wet... caveat: No code. Policy/engineering document only.	medium - second precedent, and structurally different from Mendocino (groundwater...	200
CW3E FIRO programme portal https://cw3e.ucsd.edu/firo/	doc	Index of the six FIRO programmes: Howard A. Hanson Dam, Prado Dam, Russian River Watershed, Seven Oaks Dam, Willamette Valley, Yuba & Feather Watersheds. Russian River is the one described as... caveat: I fetched and read it: it references 'decision support tools' and 'forecast verification methods' but publishes NO repository links, no...	medium - authoritative scope of the real deployed programme	200
pywr/pywr (GitHub) 191★ · 2026-07-23 · GPL-3.0 https://github.com/pywr/pywr	repo	Generalised network resource-allocation model in Python/Cython. VERIFIED to include HydropowerRecorder, HydropowerTargetParameter, ControlCurveParameter, ControlCurveInterpolatedParameter and... caveat: Linear-programme core - head-dependent hydropower is a post-hoc Recorder or iterative target parameter, not a native nonlinear constraint....	high - the ControlCurve (rule curve) and Scenario (ensemble) abstractions are exactly...	200
pywr/pywr-next (Pywr 2.0, Rust) 12★ · 2026-08-26 · MIT https://github.com/pywr/pywr-next	repo	Rust rewrite of the Pywr compute core with a Python binding layer, retaining the Parameter system from Pywr 1.x. caveat: Repo description still reads 'An experimental repository exploring ideas for a major revision'. Created 2020-11, nearly six years old and...	low - not adoptable; included so the next person does not mistake activity for readiness	200
Pywr-DRB 12★ · 2026-08-04 · MIT https://github.com/Pywr-DRB/Pywr-DRB	repo	A real, maintained basin-scale reservoir-operations application built on Pywr: Delaware River Basin, NYC reservoir system, transbasin diversions, minimum-flow targets and regulatory rules (1954... caveat: Only 12 stars and heavily US-regulatory-specific. Read it for the modelling patterns (how a legal flow target becomes a constraint), not...	high - the best worked example of encoding real institutional operating rules in an open...	200

Source	Kind	What it provides	Relevance	HTTP
iRONS toolbox 11★ · 2026-04-22 · MIT https://github.com/iRONStoolbox/iRONStoolbox	repo	Python functions for reservoir simulation, forecasting and optimisation plus interactive Jupyter Notebooks (Knowledge transfer / Implementation / Case studies). Modules: res_sys_sim.py (13 KB),... caveat: environment.yml pins python=3.9, numpy<2.0, pandas<2.0, matplotlib<3.5 - will not install into a modern env unrelaxed. Scope is comparable...	high - not as a framework to adopt (AquaSync already exceeds it) but as three donor...	200
pymoo 2946★ · 2026-07-07 · Apache-2.0 https://github.com/anyoptimization/pymoo	repo	NSGA-II, NSGA-III, R-NSGA-III, MOEA/D, GA, DE, CMA-ES, PSO with a clean Problem/Algorithm API, reference directions and decomposition methods. Deps: numpy, scipy, moocore, autograd, cma, matplotlib,... caveat: Apache-2.0 makes it the only MOEA here with no licensing friction. But at 3 decision variables it is strictly worse than the grid AquaSync...	medium - the correct MOEA choice IF AquaSync raises policy dimensionality past ~6...	200
Platypus (Project-Platypus) 654★ · repo pushe · GPL-3.0 https://github.com/Project-Platypus/Platypus	repo	Python multi-objective optimisation library (NSGA-II, epsilon-NSGA-II and others). The substrate PyBorg is built on, and the optional dependency for Pywr's [opt] extra and for iRONS. caveat: GPL-3.0 is viral. The 22-month release gap means pip users run materially older code than the repo. Only 3 open issues.	low - pymoo is the better default; listed because Pywr and iRONS both reference it	200
DEAP 6436★ · 2026-04-17 · LGPL-3.0 https://github.com/DEAP/deap	repo	Distributed Evolutionary Algorithms in Python - low-level EC toolkit (toolbox, operators, statistics, multiprocessing). Deps are just numpy and moocore. caveat: Most-starred tool in this dimension but the least suited - you build the MOO machinery yourself. Skip in favour of pymoo.	low - too low-level for this job; relevant only because HydroCNHS uses it for calibration	200
MOEA Framework 361★ · 2026-01-21 · NOASSERTION (custom/other per GitHub API) https://github.com/MOEAFramework/MOEAFramework	repo	Java framework for multi-objective optimisation with 80+ test problems and, more importantly, algorithm DIAGNOSTICS - runtime metrics, hypervolume, generational distance, attainment surfaces. caveat: Java. The Reed group uses it purely for post-processing MOEA runtime dynamics (see their MM Borg Training posts), not as a modelling tool.	low - only worth it if you must defend an MOEA's convergence quantitatively; adds a JVM...	200
Borg MOEA project site site live; · NOT open source - site text verbatim: 'Borg is freely available for academic and non-commercial use... request access to its source code' http://borgmoea.org/	tool	Auto-adaptive multi-operator search, dynamic population sizing, epsilon-dominance archiving. ANSI C, runs on Unix/Linux/Windows/Mac. The de-facto standard MOEA in the water-resources literature. caveat: The non-commercial clause is a genuine legal blocker if AquaSync is ever delivered as an operational product to KSEB/KSDMA. PyBorg...	high as a WARNING - it underpins WaterPaths, PySedSim optimisation and most EMODPS...	200

Source	Kind	What it provides	Relevance	HTTP
SDDP.jl 355★ · 2026-08-25 · NOASSERTION per GitHub API (MPL-2.0 in practice - verify before shipping) https://github.com/odow/SDDP.jl	repo	Multistage stochastic dual dynamic programming as a JuMP extension - policy graphs, Markovian and stagewise-independent uncertainty, risk measures (AV@R), and simulation of trained policies. caveat: Introduces a Julia runtime alongside Python. Requires a convex, stagewise-independent problem statement, so the reservoir must be linear...	high - the mature tool for the SEASONAL foresight problem (target storage entering the...	200
do-mpc 1443★ · 2025-10-31 · LGPL-3.0 https://github.com/do-mpc/do-mpc	repo	General nonlinear MPC and moving-horizon estimation toolbox on CasADi/IPOPT, including robust multi-stage NMPC. caveat: Not water-specific. For AquaSync's linear mass balance with box + ramp constraints, the 40-line cvxpy formulation I verified (8.8...	low today, medium later - overkill for a linear reservoir balance, but the natural...	200
HydroCNHS 22★ · 2025-03-02 · BSD-3-Clause https://github.com/cylin928/HydroCNHS	repo	Semi-distributed hydrological model for coupled natural-human systems with Dam, RiverDiv, Conveying and InSitu agent APIs, plus a DEAP-based parallel calibration and uncertainty harness. Hydrology... caveat: IMPORTANT: the repo moved from philip928lin to cylin928 and the widely-cited docs URL https://philip928lin.github.io/HydroCNHS/ now...	low - the agent-based coupling solves a problem AquaSync does not have, and its...	200
WaterPaths 23★ · 2022-01-04 · none detected by GitHub API https://github.com/bernardoct/WaterPaths	repo	C++ stochastic simulation of multi-utility water supply portfolios with dynamic risk-of-failure triggers for infrastructure investment pathways under deep uncertainty. caveat: Optimisation mode requires recompiling against Borg (non-commercial licence) plus mpirun. The...	low - wrong problem (multi-decade investment planning, not release scheduling) and...	200
PySedSim 18★ · 2020-07-22 · BSD-2-Clause https://github.com/FeralFlows/pysedsim	repo	Daily-timestep river basin simulation of flow, sediment and hydropower for evaluating dam siting, design and sediment-management operations, with MOEA coupling for operating-policy discovery. caveat: Python 2 / early-Python-3 era. Not on PyPI. The immm-sfa.github.io/pysedsim docs URL is DEAD (404). Listed as a verified negative.	low - sediment-focused, dead, and AquaSync has no sediment objective	200
M3O - Multi-Objective Optimal Operations 44★ · 2019-10-18 · GPL-2.0 https://github.com/mxgiuliani00/M3O-Multi-Objective-Optim...	repo	MATLAB toolbox implementing, side by side, deterministic DP, stochastic DP, implicit stochastic optimisation, sampling SDP, fitted Q-iteration, EMODPS (evolutionary multi-objective direct policy... caveat: MATLAB (commercial-licence blocker). The CIS Lab Python port Critical-Infrastr ucture-Systems-Lab/M3Opy is an EMPTY stub with 0 stars and...	medium as a READING reference - the canonical algorithm menu for exactly this problem...	200

Source	Kind	What it provides	Relevance	HTTP
Critical Infrastructure Systems Lab - InfeRes 16★ · 2026-01-31 https://github.com/Critical-Infrastructure-Systems-Lab/InfeRes	repo	Python package that infers reservoir water extent, level and storage volume from satellite imagery. caveat: Flagged because it surfaced during this search; it belongs to the data dimension, not optimisation. Remote-sensing-derived levels carry...	high but for a DIFFERENT gap - the credible route to AquaSync's missing pre-2020...	200
Critical Infrastructure Systems Lab - VICResOpt 35★ · 2023-08-08 · none detected by GitHub API https://github.com/Critical-Infrastructure-Systems-Lab/VICResOpt	repo	Optimisation of reservoir rule curves and operating rules embedded inside the VIC hydrologic model, with an NSGA-II driver. Companion to VICRes (verified 200, 27 stars, pushed 2026-03-15), which... caveat: Adopting it means adopting VIC, a far heavier hydrologic engine than AquaSync's SCS-CN, which would not improve a basin already replayed...	low-medium - the only maintained-adjacent example of rule-curve optimisation coupled to...	200
Inflow-Prediction-Bhakra 17★ · 2021-09-10 · MIT https://github.com/Anurag14/Inflow-Prediction-Bhakra	repo	LSTM inflow forecasting and a bidirectional storage-capacity/elevation calculator for Bhakra reservoir, Satluj basin, Northern India. caveat: Dead since 2021, single-reservoir, forecasting only (no operation optimisation). Reported to substantiate the verified negative that no...	low - the closest thing to Indian reservoir-operation code that exists on GitHub, which...	200
Reed Group software index (Cornell) actively c https://reed.cee.cornell.edu/software/	doc	Authoritative curated index of the water-resources MOEA ecosystem: Borg MOEA, MOEA Framework, OpenMORDM, Hymod & HBV, Project Platypus, PySedSim, SALib, WaterPaths, Pywr-DRB, statemodify. caveat: Marks everything 'Active' including WaterPaths (last push 2022-01) and PySedSim (2020-07), so its status labels are NOT reliable - verify...	medium - the fastest way to see the whole Reed/Cornell ecosystem and confirm nothing was...	200
Water Programming blog (Reed group) ongoing; p https://waterprogramming.wordpress.com/	doc	Long-running practitioner blog on MOEA diagnostics, Borg/PyBorg setup, MOEA Framework post-processing, many-objective reservoir optimisation and HydroCNHS walkthroughs. The practical how-to layer...	medium - the PyBorg setup post is where the licensing gate is spelled out plainly, and...	200
baobabsoluciones/flowing-basin 5★ · 2025-02-17 · MIT https://github.com/baobabsoluciones/flowing-basin	repo	Intraday multi-reservoir hydropower economic optimisation comparing RL against MILP and heuristic baselines - code behind 'Deep Reinforcement Learning for Intraday Multireservoir Hydropower...' caveat: 5 stars, 1 fork, no repo description. Reported to substantiate the verified negative that RL for reservoir operation has no...	low - the single most substantial RL-for-reservoirs repo I could find, and it has 5...	200

Verification notes

- This domain has NOT been through the adversarial verification pass. URLs carry the researching agent's own reported status only. Treat as provisional.

- dead end: Borg MOEA source code cannot be obtained by download at all. borgmoea.org gates it behind a Google form ('request access to its source code', ~3 business days, private BitBucket) and licenses it for 'academic and non-commercial use' only. This transitively poisons WaterPaths (needs a Borg recompile for optimisation mode), PySedSim's optimisation path, PyBorg, and essentially every...
- dead end: Reinforcement learning for reservoir operation - four separate GitHub searches produced nothing usable. The query 'reservoir+operation+DDPG' returned ZERO repositories. Best results anywhere: wloss/HydroPower (4 stars, dead 2020-06), Fraktal-Norge-AS/HydroRL (2 stars, 2023), vaishsRP/ReinforcementLearning-HydroelectricDam (2 stars), nanu1605/reservoir-rl (0 stars), Sarmadzandi...
- dead end: WaterPaths - do not invest. Last push 2022-01-04 (4.5 years stale), 23 stars, C++, no licence file, and it solves multi-decade multi-utility water-supply INVESTMENT pathways under deep uncertainty rather than release scheduling. Optimisation mode additionally requires recompiling against Borg plus an MPI toolchain. Its widely-linked training page...
- dead end: PySedSim - dead since 2020-07-22, Python 2 era, not on PyPI, and sediment/dam-siting focused. AquaSync has no sediment objective. Its docs site (immm-sfa.github.io/pysedsim) 404s. Not worth an afternoon.
- dead end: M3O - MATLAB, GPL-2.0, dead since 2019. Its CIS Lab Python port, Critical-Infrastructure-Systems-Lab/M3Opy, is an EMPTY stub: 0 stars, no description, no language detected. Read M3O for the algorithm menu; do not attempt to run or port it.
- dead end: Indian institutional reservoir-operation software does not exist as open source. Four search angles (IIT/NIH Roorkee tools, India-WRIS, CWC decision-support systems, named Indian reservoirs) found no NIH Roorkee or IIT hydrological package on GitHub - github.com/IIT-Roorkee and the other IIT Roorkee orgs are student robotics/CS/vision groups. India-WRIS and CWC publish data portals and...
- dead end: The FIRO programme itself publishes no software. I fetched cw3e.ucsd.edu/firo/ directly: it lists six programmes and mentions 'decision support tools' and 'forecast verification methods' but contains ZERO repository links and no technical implementation detail (contact given is kvm@ucsd.edu). Do not go looking for an official FIRO codebase - all reusable code is academic (Herman group...
- dead end: pywr-next / Pywr 2.0 - still explicitly experimental after nearly six years (repo created 2020-11). All four GitHub releases are pre-release betas and the description still warns it is exploratory. 12 stars, 54 open issues. Building on it now guarantees rework.
- dead end: swmm_mpc - dead since 2019-09 and SWMM/EPA stormwater-specific (urban drainage, pumps and orifices), not reservoir operation. It surfaces in every 'MPC water' search but does not transfer. HEC-ResSim is likewise a dead end for reuse: closed-source, Java/Windows, and its own documentation states it has no optimisation framework (v4.1 released April 2026).
- dead end: Several authoritative pages block automated verification and should not be cited as links: hec.usace.army.mil (HEC-ResSim) returns curl status 000 - connection refused to non-browser clients; escholarship.org, ascelibrary.org, sciencedirect.com, agupubs.onlinelibrary.wiley.com, iwaponline.com, mdpi.com and zenodo.org all return 403 to curl. Cite the DOI or a publisher-independent copy...

7 · Open Source River Routing, Hydraulics And 2D Flood Inundation (Plus The Dem / Hydrography / Gauge Data Supply Chain Needed To Drive Them) For The Periyar Basin

Confirmed sources (30)

Source	Kind	What it provides	Relevance	HTTP
CWC River Water Level (Manual - Hourly) - basin 031 West flowing rivers Tadri to Kanyakumari records ru · Other (Open) per CKAN license_id 'other-open'; publisher CWC / National Water Informatics Centre https://nwdp.nwic.gov.in/dataset/d951a09c-6cf8-470e-be77-...	dataset	Hourly manual river stage AND reservoir level CSVs. Basin-031 file 1991-2020 = 260 MB / 1,284,651 rows; 2021-2025 = 234 MB / 1,139,984 rows. Columns: Station, River, Basin, Latitude, Longitude,... caveat: No login, but you MUST find it by basin code 031, not by state - searching 'Kerala' misses it entirely. Cadence is nominally hourly at...	high - this single dataset closes the Muskingum-calibration gap (two synchronous hourly...	200
CWC River Discharge (Manual - Daily) - Kerala 2025-12-31 · Other (Open); publisher CWC https://nwdp.nwic.gov.in/dataset/river-discharge-manual-d...	dataset	Daily gauged discharge, 52 state resources. Kerala 2001-2025 file = 39.4 MB, 217,538 rows, 40 stations. Three on the Periyar: NEELEESWARAM 8,478 values 2001-01-01..2025-12-31; ARANGALI 7,518 values;... caveat: Manual observer record: roughly one missing day per week year-round, and a 7-day hole 2018-08-16..2018-08-22 straight across the flood...	high - the discharge half of the rating curve AquaSync needs to turn hourly Periyar...	200
CWC Study Report: Kerala Floods of August 2018 (Hydrology (S) Dte) published · Government of India report, freely hosted by KSDMA; no explicit open licence stated https://sdma.kerala.gov.in/wp-content/uploads/2020/08/CWC...	doc	5.7 MB PDF, Sept 2018. Section 4.1 is a Periyar-specific water balance: catchment areas (Neeleeswaram 4,033 km2, Mullaperiyar 637, Idukki free 605, Idamalayar 472), 15-17 Aug rainfall/runoff table,... caveat: The hourly Idukki/Idamalayar/Neeleeswaram series exist only as chart images (Figs 3, 4, 5) - the axis labels survive pdfotext but the...	high - the single most quotable ground-truth document for a 2018 replay scenario, and it...	200
NOAA-OWP/t-route 55★ · pushed_at · Apache-2.0 https://github.com/NOAA-OWP/t-route	repo	Tree-based hydrologic/hydraulic routing: Muskingum-Cunge (per-segment, parallel) and Diffusive Wave (per-subnetwork, serial), Python orchestration over Fortran kernels, CLI and BMI interfaces. caveat: SETUP COST: HIGH. No native Windows - README directs Windows users to install WSL2 with Ubuntu 20.04/22.04 first. Fortran toolchain +...	medium - the reference operational Muskingum-Cunge implementation, but it duplicates...	200

Source	Kind	What it provides	Relevance	HTTP
rileyhailes/river-route 4★ · pushed 202 · BSD-3-Clause-Clear https://github.com/rileyhailes/river-route	repo	Pure-Python vectorised Muskingum-family routing (Muskingum, RapidMuskingum, UnitMuskingum) on vector stream networks using numba kernels and scipy sparse matrices. Params file is a parquet with... caveat: NO calibration module - I checked the docs and the file-schema page; K and x are inputs you must supply from elsewhere. Only 4 stars,...	medium - cheapest possible sanity check on AquaSync's own Muskingum implementation and...	200
mhpi/dMC-Juniata-hydroDL2 (differentiable Muskingum-Cunge) 11★ · last push · CC-BY-4.0 (an unusual choice for code) https://github.com/mhpi/dMC-Juniata-hydroDL2	repo	PyTorch + Hydra implementation of Bindas et al. 2024 WRR: Muskingum-Cunge with an embedded NN mapping reach attributes (catchment area, sinuosity, slope) to Manning's n and channel geometry, trained... caveat: RESEARCH-ONLY, paper replication. Hard-wired to the Susquehanna/Juniata; no bundled data; no packaging as a library; abandoned for 2+...	low as code, high as a method to copy - learning K and x from reach attributes is...	200
delta-HBV2 / delta-MC2 global streamflow, 2.94M rivers 1980-2020 dataset v5 · Penn State Non-Commercial Software License Agreement on the dataset - commercial use requires written PSU consent https://www.nature.com/articles/s41467-025-64367-1	dataset	Nature Communications 2025 paper behind the global differentiable HBV + differentiable Muskingum-Cunge run. Companion dataset on Zenodo record 20684429 (v5, 2026-06-18): 130.9 GB in 8 continent... caveat: Zenodo returned 403 to curl from this network (anti-abuse throttle, not a dead link) - it resolves fine in a browser. Discharge only: the...	medium-high - a plausible model-based 2018 (and 1980-2020) discharge series for Periyar...	200
Deltares/SFINCS (+ hydromt_sfincs) 83★ · pushed 202 · GPL-3.0 for source; precompiled executables under Deltares Freeware (commercial + research use allowed, no redistribution/modification) https://github.com/Deltares/SFINCS	tool	Reduced-complexity 2D compound-flood solver (SFINCS-LIE drops advection; SFINCS-SSWE keeps it). Handles fluvial + pluvial + tidal + wind/wave in one run with subgrid bathymetry, and accepts... caveat: The release tags carry no GitHub assets - binaries come from download.deltares.nl or Docker Hub, so pin your version manually. Docs give...	high - the recommended path to close the 2D-inundation gap for Aluva and the Kochi...	200
LISFLOOD-FP 8.x (SEAMLESS-WAVE / Sheffield) 8.2 paper · MUDDLED - the project page says 'GNU GPL v3.0 for any non-commercial use', the 8.0 Zenodo record (4073011, 2020-10-09) says 'GNU General Public License v2.0 only', and the 8.2 paper says GPL v3.0. Get this in writing before any commercial or KSEB deployment. https://www.seamlesswave.com/LISFLOOD8.0.html	tool	The canonical academic 2D floodplain code. 8.0 added DG2/FV1 solvers; 8.1 added GPU-accelerated uniform and non-uniform ACC solvers; 8.2 (GMD 18, 9827, 2025) added GPU multiwavelet DG with dynamic... caveat: SETUP COST: HIGH. CMake + C++/CUDA build, and the 8.2 benchmarks ran on an NVIDIA A100 80 GB - the reported 1.1x-4.5x adaptivity speedups...	medium - defensible and well-published, but it buys AquaSync accuracy it cannot yet...	200

Source	Kind	What it provides	Relevance	HTTP
jsosa/LFPtools 40★ · last push · BSD-3-Clause https://github.com/jsosa/LFPtools	repo	Python CLI suite to prepare LISFLOOD-FP inputs from global data: lfp-getdepths (three methods incl. hydraulic geometry and simplified Manning), lfp-getwidths, lfp-getbankelevs, lfp-fixelevs,... caveat: Effectively abandoned; expect Python-2-era and GDAL-version friction. Use the Docker image, not a fresh pip install. A LISFLOOD-FP 5.9 BMI...	low-medium - the standard MERIT-Hydro-to-LISFLOOD bridge, and its bank-elevation and...	200
gpt-cmdr/ras-commander 76★ · pushed 202 · MIT https://github.com/gpt-cmdr/ras-commander	repo	Python API for HEC-RAS 6.x: parallel plan execution, project/plan/geometry manipulation, and - crucially - standalone HDF result reading. HdfResultsMesh exposes 2D mesh depths, velocities and WSE,... caveat: RUNNING simulations needs a real HEC-RAS install on Windows - Linux is not supported for execution. Declared version support is 3.1, 4.1,...	medium - the standalone HDF reader means AquaSync could consume someone else's...	200
psu-efd/pyHMT2D 127★ · pushed 202 · MIT https://github.com/psu-efd/pyHMT2D	repo	Control and semi-automation of 2D hydraulic models plus pre/post-processing. Supports SRH-2D, HEC-RAS and Backwater-1D; ras-commander credits it as the origin of its 2D data handling. caveat: Same fundamental constraint as ras-commander: it drives HEC-RAS/SRH-2D, it does not replace them, so you still need the Windows binaries.	medium - the multi-model option if you ever need SRH-2D as well, and the more...	200
HEC-RAS (USACE) downloads 7.0.1 repo · US public domain, free of charge https://www.hec.usace.army.mil/software/hec-ras/download...	tool	Free 1D/2D hydraulic modelling suite. Reported current release 7.0.1 (2026-06-02, ~171 MB setup, bug-fix over 7.0); a HEC-RAS 2025 Beta 0.1.0.2965 patch reportedly issued 2026-04-17. Windows 10/11... caveat: DO NOT report this URL as live without re-testing from the deployment network - I could not reach the host at all. The version numbers...	medium - the model KSEB/KSDMA consultants and the published Periyar literature actually...	000 (connection refused / unreachable from this Indian network; DNS resolves to 155.83.205.15 but both HTTPS and HTTP fail, and WebFetch got ECONNREFUSED)
Deltares/pyflwdir 116★ · pushed 202 · MIT https://github.com/Deltares/pyflwdir	repo	Pure-Python numba-accelerated terrain analysis: flow direction from elevation, upstream accumulation, Strahler/classic stream order, basin and Pfafstetter subbasin delineation, flow-direction... caveat: HAND is a terrain index, not hydrodynamics: no momentum, no backwater, no timing - so it cannot represent the tidal backwater at Aluva...	high - the cheapest credible half-step toward 2D. HAND + a stage-discharge relation...	200

Source	Kind	What it provides	Relevance	HTTP
Landlab (OverlandFlow component) 437★ · pushed 202 · MIT https://landlab.csdms.io/	tool	Python earth-surface modelling framework; the OverlandFlow component implements the de Almeida et al. (2012) local-inertial approximation of the 2D shallow water equations on a raster grid,... caveat: The user guide states outright that the algorithm 'is most stable in flat environments' and that 'instabilities can arise when trying to...	low-medium - a pip-installable 2D solver with no compiler, but the wrong stability...	200
ANUGA (anuga-community/anuga_core) 53 on the maintained anuga-community fork; 227 on the older GeoscienceAustralia/anuga_core (last push 2026-06-22)★ · anuga-comm · reported as 'Other' by the GitHub licence API (not a clean SPDX identifier) - verify before commercial use https://anuga.readthedocs.io/	tool	Finite-volume unstructured-triangle 2D shallow water solver for dam-break, riverine and tsunami inundation, in Python with C kernels. caveat: Two live forks with different star counts is a real governance smell - use anuga-community, not GeoscienceAustralia. Unstructured meshing...	low-medium - genuinely open and maintained, and unstructured meshes suit an estuary, but...	200
open TELEMAC-MASCARET actively d · GPLv3 (BIEF library under LGPL) - see https://www.opentelemac.org/index.php/licence (200) https://www.opentelemac.org/index.php/download	tool	EDF-led CFD suite; TELEMAC-2D solves depth-averaged Saint-Venant on unstructured meshes. Consortium-governed open source since 2010. caveat: SETUP COST: VERY HIGH. Fortran toolchain, MPI, and a mesh/GIS pre-processing chain (SALOME-HYDRO is Debian-only). Weeks, not days. Do not...	low - overkill for AquaSync's question, and the heaviest setup cost in this list	200
Iber (+ IberSWMM+) IberSWMM+ · free of charge from iberaula.es; NOT a recognised open-source licence - source is not openly distributed https://www.iberaula.es/space/54/downloads	tool	2D depth-averaged shallow-water model with hydrodynamics, sediment transport, water quality and rainfall-runoff; a C++/OpenMP/CUDA GPU build reports up to ~200x over the sequential version for large... caveat: 'Free' is not 'open source' - you get executables, not code, so it cannot be embedded in an AquaSync deployment the way SFINCS or...	low-medium - free, GUI-driven, GPU-capable, and the SWMM coupling is the natural route...	200
NOAA-OWP/inundation-mapping (HAND-FIM) 129★ · pushed 202 · Apache-2.0 https://github.com/NOAA-OWP/inundation-mapping	repo	The operational HAND-based Flood Inundation Mapping framework: derives HAND from a DEM, builds reach-averaged synthetic rating curves, and converts a discharge forecast to an inundation raster at... caveat: Hard-wired to US inputs: NHDPlus V21 hydrofabric, NWM reach IDs, US HUC8 tiling, USGS 3DEP DEMs. Porting to the Periyar means substituting...	medium - the reference implementation of the exact discharge-to-extent trick AquaSync...	200

Source	Kind	What it provides	Relevance	HTTP
FABDEM V1-2 (Forest And Buildings removed Copernicus DEM) V1-2 is th · CC BY-NC-SA 4.0 - NON-COMMERCIAL. Commercial queries go to fabdem@fathom.global https://research-information.bris.ac.uk/en/datasets/fabde...	dataset	Global 1 arc-second (~30 m) bare-earth DEM produced by machine-learning removal of tree and building height bias from Copernicus GLO-30. Published behind Hawker et al., Environ. Res. Lett. 17,... caveat: TWO blockers: the NC clause rules out any commercial KSEB deployment without a Fathom licence, and I could not reach data.bris.ac.uk at...	high - for floodplain hydraulics a bare-earth DEM matters more than raw resolution, and...	200
Copernicus DEM GLO-30 2024_1 rel · free and open (ESA/Copernicus licence, permits commercial use) - unlike FABDEM https://registry.opendata.aws/copernicus-dem/	dataset	Global 30 m DSM with flattened water bodies, consistent river flow direction, and edited shorelines. Free and open worldwide. Mirrored on AWS Open Data (S3, no egress cost) and on OpenTopography. caveat: It is a DSM, not bare earth - tree canopy and buildings are in the surface, which will over-constrain floodplain conveyance along the...	high - the pragmatic default DEM for a first SFINCS build: openly licensed, no...	200
MERIT Hydro / MERIT DEM v0.7 is th · free for research; registration required https://global-hydrodynamics.github.io/MERIT_Hydro/	dataset	90 m globally consistent hydrography built on the error-corrected MERIT DEM: flow direction, flow accumulation, river width, upstream area, upstream gradient, and HAND - explicitly designed for... caveat: 90 m is coarse for street-level Kochi flooding - use it for network topology, reach geometry priors (width, slope) and Muskingum K...	high - the standard input to LFPtools, hydromt_sfincs and MERIT-Basins; the pre-computed...	200
MERIT-Basins (reachhydro.org) v1.0 built · CC BY-NC-SA 4.0 - non-commercial https://www.reachhydro.org/home/parameters/merit-basins	dataset	Global vector hydrography: ~2.94M river reaches plus unit catchments, derived from MERIT-Hydro v0.7. Files named riv_pfaf_XX_MERIT_Hydro_v07_Basins_v01.shp and cat_pfaf_XX_... . Attributes include... caveat: NC licence again - a recurring theme across the best hydrography (MERIT-Basins, GRFR and FABDEM are all NC). Channel WIDTH is not an...	high - gives AquaSync a ready-made reach network with the length and slope a...	200
GRFR - Global Reach-level Flood Reanalysis covers thr · CC BY-NC-SA 4.0 (UCSD provisions) https://www.reachhydro.org/home/records/grfr	dataset	3-hourly and daily river discharge for the same 2.94M MERIT-Basins reaches, 1980-2019, with extracted flood events using a 2-year return-flow threshold. Citation Yang et al. 2021 BAMS. caveat: Naturalised reanalysis - it does NOT know about Idukki, Idamalayar or Mullaperiyar operations, so it will badly misrepresent regulated...	medium-high - an immediately usable 3-hourly modelled 2018 Periyar hydrograph at reach...	200

Source	Kind	What it provides	Relevance	HTTP
HydroSHEDS / HydroRIVERS v1.1 GeoTI - free for scientific, educational AND commercial use with attribution - the cleanest licence of any hydrography in this list https://www.hydrosheds.org/products/hydrorivers	dataset	Global river network line shapefiles at 15 arc-second resolution, distributed by continent (Asia covers India), with HydroBASINS catchments and HydroATLAS attributes alongside. caveat: Coarser and less hydrodynamically tuned than MERIT-Basins - fine for maps and basin delineation, weaker for reach geometry. Download by...	medium - the only major hydrography here with a licence permitting commercial use, which...	200
ISRO Bhoonidhi (NRSC) - CartoDEM access Bhoonidhi - EULA acceptance required at registration; government users (ministries, PSUs, government R&D; and academic institutions) get data without further clearance. Sub-30 m products are commercial via NSIL. https://bhoonidhi.nrsc.gov.in/bhoonidhi/registration.html	dataset	ISRO's current EO data hub and the live route to CartoDEM (Cartosat-1 derived Indian national DEM). CartoDEM at 30 m and coarser is free; the archive documents 1/3 arc-sec source subsampled to 1 and... caveat: Registration-gated, tile-by-tile, no scriptable API - budget half a day of clicking for the Periyar corridor. CartoDEM v1 vertical...	medium - the politically and institutionally right DEM for an India-deployed system, and...	200
GHI / GHSA - Geospatial dataset for Hydrologic analyses in India / South Asia GHSA v2504 - CC BY 4.0 - commercially usable, unlike most of the hydrography above https://essd.copernicus.org/articles/15/4389/2023/	dataset	Quality-controlled CWC/WRIS gauge metadata, catchment boundaries and hydrometeorological time series. v1 (ESSD 2023): 472 vetted stations, Peninsular India, water years 1950-2020. v2 (Zenodo... caveat: IMPORTANT LIMITATION I verified in the readme: streamflow is distributed at ANNUAL and MONTHLY resolution only (flow_mcm_obs,...	medium - the fastest way to get a QC'd, relocated gauge location and delineated...	200
GUARDIAN - sub-daily river discharge over India code repo - paper CC BY-NC-ND 4.0; data on figshare doi 10.6084/m9.figshare.27004282 (verified, resolves 202) https://pmc.ncbi.nlm.nih.gov/articles/PMC11489425/	paper	Scientific Data 2024. Extends India-WRIS stage into sub-daily discharge at 210 gauging stations: hourly during the monsoon (June-Sept), sub-daily otherwise, 2020 onward, by fitting rating curves... caveat: The paper never confirms Neeleeswaram is among the 210 stations - the station list is a figure, not a parseable table. Coverage starts...	high as METHOD - precisely the recipe AquaSync should run on Neeleeswaram to convert its...	200
Global Flood Database v1 (2000-2018) v1 ends 20 - open via the Earth Engine catalogue https://developers.google.com/earth-engine/datasets/catalog...	dataset	913 mapped flood events from MODIS at 250 m, each image giving maximum flood extent, inundation duration and cloud-observation quality, filterable by date, country or Dartmouth Flood Observatory ID.... caveat: 250 m MODIS under monsoon cloud is a weak validation target - persistent cloud on 15-17 Aug 2018 means the max-extent layer is partly...	medium - an OBSERVED 2018 inundation extent to validate any 2D model AquaSync eventually...	200

Source	Kind	What it provides	Relevance	HTTP
ec-jrc/lisflood-code (LISFLOOD-OS, the JRC hydrological model) 167★ · pushed 202 · EUPL-1.2 - a genuine open-source licence, commercially usable https://github.com/ec-jrc/lisflood-code	repo	Spatially distributed rainfall-runoff-routing model in Python, the engine behind EFAS and GloFAS v3.1+ and the European Drought Observatory. Represents snow, interception, infiltration, groundwater,... caveat: Confusing the two LISFLOODs is the most common error in this literature. LISFLOOD-FP = Bristol/Sheffield 2D floodplain hydraulics....	low-medium - NAME COLLISION WARNING: this is NOT LISFLOOD-FP. It is catchment hydrology,...	200

Verification notes

- This domain has NOT been through the adversarial verification pass. URLs carry the researching agent's own reported status only. Treat as provisional.
- dead end: indiawris.gov.in is unreachable (curl 000, repeated, on both bare and /wris paths) from this network, and multiple sources report that daily discharge for some stations there is classified behind a login. Do NOT route AquaSync's gauge ingestion through India-WRIS. Use nwdp.nwic.gov.in (National Water Data Portal, CKAN, verified 200, no login, working /api/3/action/package_search and...
- dead end: GHI / GHSA (Zenodo, CC BY 4.0) looks like the obvious Indian streamflow dataset and is NOT a hydrograph source. I read its readme: streamflow is distributed at ANNUAL and MONTHLY resolution only (flow_mcm_obs, flow_mcm_tot), with days_avail/days_tot merely counting how many daily observations existed upstream. Daily series are not in the package. Use it for QC'd station coordinates and...
- dead end: CWC 'River Discharge (Telemetry - Hourly)' on NWDP has only 14 basin resources and basin 031 (West flowing rivers Tadri to Kanyakumari) is NOT among them. There is no public hourly discharge for the Periyar anywhere. Stop looking - build the rating curve from hourly stage plus daily discharge instead.
- dead end: The 8,800 cumec / 16 Aug 2018 14:00 Neeleeswaram peak cannot be sourced from any machine-readable public record. The NWDP daily discharge series has a 7-day hole from 2018-08-16 to 2018-08-22, and the CWC report carries it only as a chart image (Fig. 3) whose axis labels survive pdftotext but whose data points do not. Either digitise the figure or derive it from the hourly stage.
- dead end: The Idukki and Idamalayar reservoir level series on NWDP start 2019-06-10, NOT 2018. They are genuinely valuable (they pre-date the amith-vp feed by 14 months and span its corrupt window) but they do NOT close the 2018 reservoir-operations gap. Only the CWC report narrative covers 2018 Idukki/Idamalayar operations.
- dead end: www.hec.usace.army.mil is completely unreachable from this network - curl returns 000 on both HTTPS and HTTP after DNS resolves to 155.83.205.15, and WebFetch got ECONNREFUSED. Every HEC-RAS version claim in circulation (7.0.1, 2026-06-02, 171 MB) therefore comes from search snippets, not a fetched page. Assume US .mil geoblocking and plan for a VPN, or drop HEC-RAS from the execution...
- dead end: data.bris.ac.uk - the actual FABDEM file host - is unreachable from here (curl 000 in 0.67 s despite DNS resolving to 137.222.0.78; data.bristol.ac.uk also 000). Only the Bristol research-information landing page responds. Combined with FABDEM's CC BY-NC-SA 4.0 licence (commercial queries go to fabdem@fathom.global), do not assume FABDEM is available for a KSEB deployment until both...
- dead end: MERIT Hydro cannot be scripted: access requires completing a Google Form and waiting for a download password by email. Budget a day of latency, not a curl command. MERIT-Basins and GRFR are distributed via Google Drive and Globus - also not curl-able.
- dead end: The OpenTopography programmatic DEM endpoint (portal.opentopography.org/API/globaldem) returns 401 to an unauthenticated request - a free API key is mandatory. Only the interactive /raster UI works

without one. Register before writing any DEM-fetch automation. Separately, I found NO Copernicus EMS EMSR activation for Kerala August 2018 - India went through the International Charter...

- dead end: Zenodo consistently returns 403 to curl from this IP ('Access to this resource has been restricted due to unusual traffic'), including record and API endpoints for GHI, LISFLOOD-FP and the delta-MC2 global dataset. These are NOT dead links - all resolve normally via browser/WebFetch. Any status-checking script should treat zenodo.org 403 as 'reachable, throttled' rather than broken....

8 · State Of The Art In Dam Operation Decision Support: What Is Deployed, What Fails

Confirmed sources (30)

Source	Kind	What it provides	Relevance	HTTP
CAG Report No. 6 of 2021 - Performance Audit: Preparedness and Response to Floods in Kerala presented · Government of India audit report, public https://cag.gov.in/uploads/download_audit_report/2021/Ker...	standard	The single most valuable document found. Official audit with IISc Bangalore as technical consultant. Contains: the exact 1983 and 2020 rule curves for Idukki and Idamalayar (Appendix 3.3, 10-day... <i>caveat: Appendix 3.2 (1983 rule curve) is a scanned image with no extractable text. The daily appendix tables are simulations, but the inflow /...</i>	high - closes AquaSync's no-2018-data gap, supplies the official rule curve the...	200
CWC Study Report: Kerala Floods of August 2018 (Hydrology (S) Directorate) September · Government of India, public https://sdma.kerala.gov.in/wp-content/uploads/2020/08/CWC...	standard	Primary official 'dams did not cause it' analysis. Idukki: 731.82 m on 10 Aug (0.61 m below FRL, only ~40 MCM cushion), drawn to 730.80 m by 13 Aug (~90 MCM), peak inflow 2532 cumec at 22:00 on 15... <i>caveat: CWC's Idamalayar peak inflow (1164 cumec) directly contradicts KSEBL's 2328 cumec quoted in the CAG report. Do not treat either as ground...</i>	high - the authoritative baseline hydrology for the Periyar in Aug 2018 and the origin...	200
Sudheer et al. (2019), Role of dams on the floods of August 2018 in Periyar River Basin, Kerala, Current Science 116(5):780-794 10 March 2 · Current Science, freely downloadable https://www.currentscience.ac.in/Volumes/116/05/0780.pdf	paper	The HEC-HMS study the Kerala government relies on. Twelve scenarios of reservoir storage (85/75/50/25%) at end-June and end-July 2018. Only 16-21% peak attenuation from advance emptying; below 50%... <i>caveat: Methodologically attacked by J. Harsha (CWC) for SCS-CN with lambda=0.2 untested for the Periyar and 2010 land cover extrapolated to 2018....</i>	high - defines the realistic upper bound on what any reservoir-operation optimiser can...	200
Mishra et al., The Kerala flood of 2018: combined impact of extreme rainfall and reservoir storage (HESS Discussions preprint hess-2018-480) posted 14 · CC-BY (Copernicus preprint) https://hess.copernicus.org/preprints/hess-2018-480/	paper	The preprint the court-appointed Amicus Curiae relied on. Finds 6 of 7 major reservoirs above 90% capacity on 8 Aug 2018 and >500-year return-period rainfall upstream of Idukki, Kakki and Periyar.... <i>caveat: NOT PEER-REVIEWED AND NOT PUBLISHED. Copernicus status: 'The manuscript was not accepted for further review after discussion.' Any...</i>	medium - historically pivotal but must be cited with its status, never as a...	200
SANDRP guest article: Role of dams in Kerala floods - Distortion of Science (J. Harsha, Director, CWC Chennai) 25 August · SANDRP, freely readable https://sandrp.in/2020/08/25/role-of-dams-in-kerala-flood...	doc	The technical rebuttal Current Science refused to publish. Attacks Sudheer et al.'s SCS-CN parameterisation (lambda=0.2 taken from USDA and never validated for the Periyar), the use of 2010... <i>caveat: Advocacy-site publication, not peer-reviewed - because Current Science declined it without peer review. Author is a serving CWC officer...</i>	high - the specific criticisms land squarely on the SCS-CN rainfall-runoff method...	200

Source	Kind	What it provides	Relevance	HTTP
SANDRP: Response of Sudheer et al. to J. Harsha's comments, with Harsha's rejoinder 19 Septemb · SANDRP https://sandrp.in/2020/09/19/respons-e-of-sudheer-et-al-to-...	doc	The other half of the exchange - the IIT Madras team's defence of their assumptions and Harsha's reply. The only place the full argument appears in one thread. caveat: Same provenance caveat as the Harsha critique.	medium - needed to represent the disagreement fairly rather than quoting only the attack	200
Vegad & Mishra (2025), Climate change and effectiveness of dams in flood mitigation in India, npj Natural Hazards published · open access (npj) https://www.nature.com/articles/s44304-025-00117-z	paper	178 major Indian dams, in-situ plus satellite observations with hydrological and hydrodynamic simulation. Dam effectiveness in flood mitigation depends MORE on antecedent reservoir storage than on... caveat: National-scale, not Periyar-specific. Same lead-author group as the rejected 2018 preprint, so expect the Kerala government to discount it.	high - the strongest recent peer-reviewed support for AquaSync's premise, from the same...	200
Suresh et al. (2024), Satellite-based Tracking of Reservoir Operations ... 2018 Extreme Weather Event in Kerala (Remote Sensing of Environment) - open NASA NTRS manuscript revised 26 · NASA NTRS open access (accepted manuscript) https://ntrs.nasa.gov/api/citations/20240007715/downloads...	paper	RAT 3.0 applied to 19 Kerala reservoirs for 2018 with verified inflow skill: Idamalayar r=0.81 / RMSE 135.29 cumec / NRMSE 13.30%; Kakki r=0.83 / 83.54 / 11.19%; Banasurasagar r=0.78 / 31.81 / ... caveat: The ScienceDirect version of record is paywalled (400/403 to automated fetch). This is the accepted manuscript, so page and figure numbers...	high - independent quantified evidence of how hard Idukki inflow estimation is, and the...	200
Das et al. (2025), Forecast-Informed Reservoir Operations within a Satellite-Based Framework for Mountainous and High-Precipitation Regions: Case of the 2018 Kerala Floods, J. Hydrologic Engineering 30(2) April 2025 · paywalled (ASCE) https://ascelibrary.org/doi/10.1061/JHYEFF.HEENG-6276	paper	Published prior art on essentially AquaSync's thesis: forecast-informed reservoir operation applied to the 2018 Kerala flood using GEFS precipitation forecasts within the RAT satellite framework.... caveat: ascelibrary.org returns 403 to curl AND WebFetch - I verified existence, authorship, journal and date via Crossref only. I have NOT read...	high - AquaSync must position against this, not around it; it is the closest published...	403
KSDMA Dam Water Level portal (daily KSEB and Irrigation bulletins) updated da · Government of Kerala, public https://sdma.kerala.gov.in/dam-water-level/	dataset	Stable entry point to the official daily reservoir bulletin. Links the current KSEB PDF and Irrigation PDF plus a Malayalam explanatory note on reading the tables. Live and current - the page... caveat: PDF only - no API, no JSON, no historical archive on the page; older bulletins survive only as orphaned wp-content URLs. Column layout is...	high - authoritative daily source for rule level and blue/orange/red alert levels, which...	200
KSDMA daily KSEB dam bulletin, 25 August 2026 edition 25 August · Government of Kerala, public https://sdma.kerala.gov.in/wp-content/uploads/2026/08/KSE...	dataset	Verified example of the schema: FRL, today's water level, rule level (ft/m), Blue (m), Orange (m), Red (m), total live storage at FRL (MCM), live storage (MCM), percentage storage, current spillway... caveat: Rolling filename - this specific URL will be superseded; harvest via /dam-water-level/ daily. Note the KSEB total live storage at FRL for...	high - independently confirms the CAG 2020 rule-curve values are in force today, and...	200

Source	Kind	What it provides	Relevance	HTTP
KSEB Limited Dam Safety Organisation KSEB, public pages https://dams.kseb.in/?page_id=39	doc	Official dam-owner site covering 59 structures including the four majors (Idukki, Idamalayar, Kakki, Banasurasagar). Download sections for Emergency Action Plans (?p=333), Monthly Statistics... caveat: No rule-curve document is linked directly from the dams page. A 'Rule Curve Major Reservoirs of KSEBL' document circulates only on Scribd,...	high - the actual dam owner and the concrete route for KSEB integration	200
Dam Safety Act 2021 (Act No. 41 of 2021) - full text assented 1 · Government of India statute, public https://wrd.maharashtra.gov.in/Site/Upload/Pdf/DSA_2021_a...	standard	Verified full text (459 KB, real PDF, correct gazette header). s.35(1)(a): every specified-dam owner shall establish a hydro-meteorological network AND AN INFLOW FORECASTING SYSTEM; (b) an emergency... caveat: A state-government mirror, not the canonical India Code copy (indiacode.nic.in handle 17002 and both bitstream PDF paths 404 as of 26 Aug...	high - simultaneously the legal mandate that justifies AquaSync and the liability...	200
Lake Mendocino FIRO Final Viability Assessment (December 2020) December 2 · public (UC San Diego / eScholarship) https://cw3e.ucsd.edu/FIRO_docs/LakeMendocino_FIRO_FVA.pdf	standard	The definitive FIRO reference. Table E.1 alternatives and median 10-May storage increase (baseline 0%, EFO 27%, Hybrid EFO 15%, Modified Hybrid EFO 20% preferred, Five-Day Deterministic 18%).... caveat: The recommended citation URL escholarship.org/uc/item/3b63q04n returns 202 (a Cloudflare interstitial, not a real failure) - use this cw3e...	high - the template for the evidence burden AquaSync would face, and the source of the...	200
Prado Dam FIRO Final Viability Assessment (December 2023) December 2 · public https://cw3e.ucsd.edu/FIRO_docs/Prado/FIRO_Prado_FVA.pdf	standard	Second completed FVA. Steering committee formed late 2017, first met March 2018, PVA 2021, FVA Dec 2023. Buffer pool raised 498 -> 505 ft in April 2021 by an interim WCM modification. 26 scenarios... caveat: Prado's benefit is groundwater recharge, not flood control or hydropower, so the transfer to a hydropower-first basin like the Periyar is...	high - shows the graduated regulatory ladder (interim WCM change -> minor deviation ->...	200
Scripps/UCSD: New forecast-informed decision-making tool implemented at Coyote Valley Dam and Lake Mendocino October 20 · news release https://scripps.ucsd.edu/news/new-forecast-informed-decis...	doc	Confirms the water control manual update was signed 22 October 2025 by USACE, Sonoma Water, Scripps/CW3E and California DWR; the first redefinition of flood-control operating rules in the dam's... caveat: News release, not the manual. I could not locate the signed WCM as a public PDF. The article contains no quotes about limitations or...	high - the single most important date in the FIRO story and the direct answer to 'what...'	200
USACE Draft EA: Water Year 2021-2026 Major Planned Deviation to the Coyote Valley Dam - Lake Mendocino Water Control Manual covers WY2 · public (NEPA document) https://www.sonomawater.org/media/PDF/Environment/Environ...	standard	The NEPA document bridging the FVA and the permanent WCM. Authorises storing an additional 11,650 acre-feet above the existing guide curve for WY2021 through WY2026, evaluating Hybrid and Modified... caveat: This is the DRAFT EA; the signed FONSI / final EA is a separate document I did not locate.	high - proves the practical route is 'prove it under a time-limited deviation, then...	200

Source	Kind	What it provides	Relevance	HTTP
Forbis et al. (2025), Application of forecast-informed reservoir operations at US Army Corps of Engineers dams in California, J. Flood Risk Management 18(1) published · open access https://onlinelibrary.wiley.com/doi/10.1111/jfr3.13051	paper	Open-access USACE-authored account. States plainly that 'implementing FIRO at any given site is accomplished through updating its WCM'. Names the barriers: forecast skill at sufficient lead time is... caveat: US-specific; says nothing about monsoon regimes or hydropower-first systems.	high - the most honest institutional account of what fails, written from inside the agency	200
Delaney et al. (2020), Forecast Informed Reservoir Operations Using Ensemble Streamflow Predictions for a Multipurpose Reservoir in Northern California, Water Resources Research September · Wiley (verify OA status) https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/...	paper	The EFO algorithm itself: each member of a CNRFC ensemble streamflow prediction is routed through the reservoir individually to compute the probability of reaching critical operational thresholds,... caveat: The 33% figure is for an optimised risk tolerance curve; the FVA's adopted Modified Hybrid EFO delivers 20%. Do not conflate them. The...	high - the concrete deployed answer to AquaSync's perfect-foresight problem: replace...	200
Brodeur, Delaney, Whitin & Steinschneider (2024), Synthetic Forecast Ensembles for Evaluating Forecast Informed Reservoir Operations, Water Resources Research February 2 · Wiley (verify OA status) https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/...	paper	Directly attacks the problem AquaSync has: you cannot evaluate a forecast-informed policy when you only hold a few years of real forecasts. Generates large synthetic ensembles of hydrologic... caveat: Both companion repos are unlicensed and unstarred - reference implementations, not maintained libraries. Expect to reimplement rather than...	high - the methodologically correct way to test AquaSync's optimiser without a long...	200
zpb4/syn-forecast_stochastic - synthetic hydrology and forecast generation with a FIRO model 1★ · pushed 202 · MIT https://github.com/zpb4/syn-forecast_stochastic	repo	The live successor to the 2024 synthetic-ensemble work. Python, MIT licensed, supporting a manuscript titled 'Forecast Informed Reservoir Operations (FIRO) for the future: Exploring the operational... caveat: 1 star, 196 KB, six months old - a manuscript companion, not a library. The associated paper appears not yet published.	high - MIT-licensed and actively developed, so unlike the other FIRO repos it can...	200
Albano et al. (2026), Characterizing Watershed Responses and Reservoir Operational Flexibilities: Analyses to Support FIRO Planning and Assessments, Water Resources Research published · open access (Wiley/AGU) https://doi.org/10.1029/2025WR042131	paper	A transferable screening method for whether FIRO is viable at a given reservoir before committing to a full viability assessment. Defines an 'operational flexibility' metric - how reservoir inflow... caveat: No data or code repository advertised. CA/NV only, and it says nothing about monsoon regimes - transfer to a tropical basin with a 3-4 day...	high - gives AquaSync a defensible, citable way to answer 'is forecast-informed...	200
F. M. Ralph, FIRO briefing to the California Water Commission, 16 April 2025 16 April 2 · public meeting material https://cwc.ca.gov/-/media/CWC-Website/Files/Documents/20...	doc	Authoritative status of every FIRO pilot as of April 2025: Howard Hanson Dam (Green River, Seattle District), Willamette Valley (14 dams, Portland District), New Bullards Bar (FVA complete),... caveat: Slides, so figures are headline numbers without methodology. Predates the October 2025 WCM signing.	high - the current programme scoreboard, and the 85%-of-variance-from-5%-of-days...	200

Source	Kind	What it provides	Relevance	HTTP
GAO-16-685: Army Corps of Engineers - Additional Steps Needed for Review and Revision of Water Control Manuals 26 July 20 · public (US GAO) https://www.gao.gov/products/gao-16-685	standard	The institutional failure mode, documented. USACE operates more than 700 dams. Engineer regulations require water control manual review no less than every 10 years, but the extent of reviews and... caveat: 403 to curl - I read it via WebFetch, not a direct download, so I could not verify page-level detail. Predates the FIRO programme's...	high - the honest answer to 'how do water control manuals get changed': mostly they do...	403
Delft-FEWS FAQ (Deltares) unknown (c · zero-cost EULA; source not public) https://oss.deltares.nl/web/delft-fews/faq	tool	Authoritative licensing statement. Delft-FEWS is available at zero licence cost - you only accept the licence agreement - but it is 'Open Software, but not Open Source': Deltares retains central... caveat: NO INDIA DEPLOYMENT IS DOCUMENTED on the Deltares pages - countries named are Sudan, Pakistan, Czech Republic, Netherlands, England,...	high - the de facto operational platform is free to licence but not forkable, which...	200
UW-SASWE/RAT - Reservoir Assessment Tool 3.0 26★ · pushed 202 · GPL-3.0 https://github.com/UW-SASWE/RAT	repo	The satellite-based reservoir monitoring framework behind the Kerala papers. Estimates surface area, storage change, inflow (VIC hydrological model), outflow, evaporation and area-elevation curves... caveat: GPL-3.0 is copyleft - linking RAT into AquaSync imposes GPL obligations on the combined work. Their own paper reports Idukki inflow...	high - a working, licensed, maintained baseline for satellite-derived Idukki/Idamalayar...	200
SASWE operational Kerala reservoir monitoring instance (RAT 3.0) live as of · unstated on the portal; underlying RAT is GPL-3.0 https://depts.washington.edu/saswe/kerala/	api	Live Kerala deployment of RAT 3.0 with per-reservoir inflow, outflow, surface area, temperature, storage change, area-elevation curve, evaporation and suspended sediment, a Forecast Date selector,... caveat: Research-lab hosted rather than an official Kerala government service, so continuity depends on grant funding. I did not verify update...	high - an existing, funded, Kerala-government-linked operational system covering the...	200
Koo et al. (2025), Balancing Operators' Risk Averseness in Model Predictive Control for Real-time Reservoir Flood Control (arXiv preprint; J. Hydroinformatics 2025) submitted · arXiv preprint; journal version J. Hydroinformatics (2025) jh2025019 https://arxiv.org/abs/2407.04506	paper	States the deployment failure directly: 'optimisation methodologies have not been widely applied in real-time operation due to disparities between research assumptions and practical requirements'.... caveat: No code availability stated. Single reservoir in a temperate East Asian monsoon - closer to Kerala than California is, but still not the...	high - AquaSync's optimiser searches over (target level, start hour, max rate) but...	200
Rayner, Lach & Ingram (2005), Weather forecasts are for wimps: why water resource managers do not use climate forecasts, Climatic Change 69(2):197-227 2005 · Springer, likely paywalled https://link.springer.com/article/10.1007/s10584-005-3148...	paper	The canonical study of why operators reject forecasts: traditional reliance on large built infrastructure, organisational conservatism and complexity, mismatch of forecast temporal and spatial... caveat: US water utilities, 2005, seasonal climate forecasts rather than short-range flood forecasts. Directionally right, specifics dated....	high - twenty years on it still explains why a technically correct optimiser gets...	200

Source	Kind	What it provides	Relevance	HTTP
Google Flood Forecasting API / Flood Hub live; meth · Google API terms; access by application https://developers.google.com/flood-forecasting	api	Operational riverine flood forecasting API: daily hydrologic forecasts to a 7-day horizon plus flood status updated several times a day, over 5,000 verified gauge locations and ~250,000 virtual... caveat: I COULD NOT CONFIRM gauge or virtual-gauge coverage on the Periyar specifically, nor whether forecasts exist upstream of Idukki where they...	medium - the most plausible free route to 7-day forecasts that would let AquaSync...	200

Verification notes

- This domain has NOT been through the adversarial verification pass. URLs carry the researching agent's own reported status only. Treat as provisional.
- dead end: indiacode.nic.in for the Dam Safety Act 2021: BOTH the handle page (/handle/123456789/17002, with and without query strings) and the bitstream PDFs (/bitstream/123456789/17002/1/A2021-41.pdf and .../aA2021-41.pdf) return 404 as of 26 Aug 2026, despite being the top search result. Use the Maharashtra WRD mirror instead - a verified real 459 KB PDF with the correct gazette header.
- dead end: damsafety.in/ecm-includes/PDFs/Dam_Safety_Act_2021.pdf returns HTTP 200 but downloads ZERO BYTES. A 200 status code is not proof of content - always check file size and file type after downloading. Same trap in reverse: egazette.gov.in/WriteReadData/2021/232128.pdf returns 200 and is a genuine 1.5 MB PDF, but its content is an NH22 land-acquisition notification, NOT the Dam Safety Act.
- dead end: publications.usace.army.mil blocks everything: er_1110-2-240.pdf and em_1110-2-3600.pdf return 403 to curl with full browser headers AND to WebFetch, from multiple district mirrors (nwd.usace.army.mil also 403). I could not directly read the USACE deviation regulations; the three-category description (planned / unplanned / emergency, with emergency approvable by phone followed by a...
- dead end: gao.gov returns 403 to curl even with a full Chrome UA and Accept headers, but WebFetch reads it fine. The exact reverse holds for sandrp.in, which returns 403 to WebFetch but 200 to curl with a browser UA. Neither tool alone is sufficient - always try both before concluding a source is unreachable.
- dead end: India-WRIS (indiawris.gov.in) is completely unreachable from this host: 000 on the root, /wris/, arc.indiawris.gov.in and a POST to /Dataset/Reservoir, with connections hanging rather than refusing. This is the obvious place to look for pre-2020 Indian reservoir time series and I could not test it at all. It may be geo-restricted to India - retest from an Indian network before...
- dead end: Mishra et al. 'The Kerala flood of 2018' (hess-2018-480) is NOT a peer-reviewed paper. Copernicus status: 'The manuscript was not accepted for further review after discussion.' Much of the 2018-2021 commentary cites it as published, including material that fed the Amicus Curiae report, and the Kerala government has used precisely that error to discredit the whole dam-mismanagement...
- dead end: J. Harsha's 'Role of dams in Kerala Floods 2018: What are we missing?' is NOT indexed in Crossref - I searched by author, by title, and within the ISH Journal of Hydraulic Engineering (ISSN 0971-5010) and found nothing. It exists only as a ResearchGate entry and as SANDRP guest articles, because Current Science declined the rebuttal without peer review. There is no citable journal...
- dead end: cw3e.ucsd.edu/firo_lake_mendocino_fva - the URL printed in the official Lake Mendocino one-page summary (February 2021) - is DEAD (404). The live FIRO hub is <https://cw3e.ucsd.edu/firo/>. Likewise github.com/hydrophile/EFO does not exist; the actual EFO code repo is [hydrophile/WRR-Manuscript-Lake-Mendocino-EFO-Model](https://github.com/hydrophile/WRR-Manuscript-Lake-Mendocino-EFO-Model) (MATLAB, 7 stars, last push 2020-04-06, 84 MB, and crucially NO LICENSE...

- dead end: escholarship.org/uc/item/3b63q04n - the recommended citation URL for the Lake Mendocino FVA - returns 202, a Cloudflare interstitial rather than a genuine failure. Do not record it as broken and do not script around it; fetch the PDF directly from cw3e.ucsd.edu. Similarly the 2019 KSDMA 'Orange Book of Disaster Management' (verified 200) is NOT the source of Kerala's dam alert levels...
- dead end: Publisher paywalls that defeat both curl and WebFetch: ascelibrary.org (403 - blocks Das et al. 2025, the closest published competitor, for which I could verify only title/authors/journal/date via Crossref), sciencedirect.com (400/403), and nature.com which issues a 303 to idp.nature.com for automated fetch. For all three the Crossref REST API (api.crossref.org/works/) reliably...

9 · Integrating With Existing Dam, Scada And Government Water Systems (Kseb / Ksdma / Cwc / Ndsa), Industrial Protocols, Open Water Data Standards, Dam Safety Act 2021 Compliance, And Cybersecurity Constraints

Confirmed sources (29)

Source	Kind	What it provides	Relevance	HTTP
KSEB Dam Safety Organisation - Monthly Statistics (daily reservoir water levels, 91 Excel workbooks) July 2026 · Government of Kerala / KSEB Ltd public download, no explicit licence stated https://dams.kseb.in/?p=329	dataset	91 monthly .xlsx/.xls workbooks 'Water level of main reservoirs of KSEB Limited', Jan 2019 to Jul 2026, one sheet per day (sometimes 3-hourly during floods). Download pattern... caveat: IDs 310-314 (Jan-May 2019) 302-redirect to http://kseboa.in/... which does not resolve (curl 000) - those five months are UNAVAILABLE....	high - authoritative primary KSEB source for Idukki and Idamalayar, starts 14 months...	200
KSEB monthly workbook, June 2019 (worked example / proof of 2018 data) published · KSEB Ltd public download https://dams.kseb.in/?wpdmdl=315	dataset	31 sheets: 'Sheet1 (2)' plus 01.06.19 to 30.06.19. Header row: SL NO RESERVOIR MWL (m) FRL (m) Crest Level (m) Present Water Level (m) Previous Year Water Level Percentage Storage ... caveat: Idukki levels are in FEET in these sheets, Idamalayar in METRES - mixed units within one row set. Kuttiyadi rows show an obvious...	high - the concrete proof that KSEB workbooks carry a same-date previous-year column,...	200
KSEB Ltd live dam water level dashboard live, serv · KSEB Ltd public web page, no licence stated https://kseb.in/dashboard/view/water-level	api	HTML table, 17 columns, all KSEB reservoirs: Today's Water level (m), Today's Live Storage (MCM), %Storage wrt Live Storage at FRL, Same day previous year Water level (m), Same day previous year... caveat: Archive is SHALLOW. Verified rows for 2026-08-01, 2026-08-20 and 2026-08-26; empty for 2026-07-01 and every earlier date I tried back to...	high - the only free source I verified that publishes the seasonally varying...	200
Kerala State Load Despatch Centre - System Statistics: Storage live, serv · KSEB/Kerala SLDC public page, no licence stated https://sldckerala.com/index.php?id=7	api	Daily reservoir table with columns MDDL (m) FRL (m) Effective Storage at FRL (mcm) Gen. Capability at FRL (mu) Reservoir Level (m) Effective Storage (mcm) Storage (%) Gen. Capability... caveat: HISTORICAL RETRIEVAL DOES NOT WORK. The page carries a POST form (action=index.php, hidden txtDate=943900200, submit name=sbtstore) and a...	high - free daily OBSERVED INFLOW for exactly AquaSync's two reservoirs, the variable...	200

Source	Kind	What it provides	Relevance	HTTP
KSDMA - Dam Water Level (daily consolidated KSEB + Irrigation bulletins) page dated · Government of Kerala https://sdma.kerala.gov.in/dam-water-level/	doc	Two daily PDFs - 'Water Levels of Major Reservoirs (KSEB)' and '(Irrigation)' - updated daily at 11:00 IST, plus a guidelines document on how to read the tables. caveat: PDF ONLY - no CSV, no JSON, no API. Requires PDF table extraction. Combines KSEB (hydel) and Irrigation Department reservoirs in separate...	high - this is the official disaster-management-side view of the same numbers, and it is...	200
CWC - Study Report: Kerala Floods of August 2018 (Hydrological Studies Organisation, Sept 2018) authored S · Government of India / CWC, published by KSDMA https://sdma.kerala.gov.in/wp-content/uploads/2020/08/CWC...	paper	48 pp, 6.0 MB, fully text-extractable. Hourly narrative of Idukki reservoir operation 09-18 Aug 2018 (levels, spill, powerhouse release, peak inflow 2532 cumec at 22:00 on 15 Aug, 41% attenuation,... caveat: The hourly inflow/outflow/level series exist only as CHART IMAGES (Fig.4, Fig.5, Fig.3) - the PDF text layer preserves the axis tick...	high - the single best verified source of 2018 Idukki/Idamalayar operational data AND...	200
CWC - Report on Kerala Flood & Solutions (December 2018) authored D · Government of India / CWC, published by KSDMA https://sdma.kerala.gov.in/wp-content/uploads/2020/10/Ker...	paper	60 pp, 9.7 MB, text-extractable. MIKE FLOOD / MIKE 11 2-D hydrodynamic modelling of Periyar, Pamba, Chalakudy and the Vembanad Lake catchment. Sec 7.1 MIKE 11 calibration at Neeleswaram. Sec... caveat: CWC explicitly notes 'the hourly dam inflow/outflow through rivers were not yet received from the project authorities' - so even CWC...	high - gives an officially published 8-hour Idukki+Idamalayar to Neeleswaram travel time...	200
Encardio Rite - Idukki Dam monitoring case study (the incumbent SHM/EWS system) undated bl · vendor marketing content, all rights reserved https://www.encardio.com/blog/iddam-monitoring	doc	Names the contract - 'Real-Time Early Warning Structure Health Monitoring and Data Interpretation (RESHMI)' under DRIP - and the stack: geotechnical + geodetic sensors (prism targets on the... caveat: Vendor marketing, not a spec sheet: gives NO instrument COUNTS, no installation date, no protocol version, no sampling rate. Cross-check...	high - defines the actual technical boundary at Idukki: there is no open protocol...	200
NDSA - Regulations notified under the Dam Safety Act 2021 (all 20) latest reg · Government of India, Official Gazette https://ndsa.gov.in/documents/dam-safety-act-regulation	standard	The complete notified regulation set with publish dates. The four that bind AquaSync: Reg 1 (13/03/2024) log books/database under s.18; Reg 5 (13/04/2024) EAP update interval under s.36(1)(b); Reg 8... caveat: The /documents/dam-safety-act-regulations spelling ALSO returns 200 but renders an empty shell; the working route is...	high - this is the statutory basis for pitching AquaSync's telemetry layer as a...	200
NDSA public document API (undocumented but open) queried li · Government of India public data, no licence stated https://ndsa.gov.in/ndsa-backend/web/api/document-config	api	GET returns the document-type map: {guidelines, reports, order-and-notice, regulation} with frontendRoute/apiKey/apiEndpoint. Then POST https://ndsa.gov.in/ndsa-backend/web/regulation?lang=en with... caveat: UNDOCUMENTED and reverse-engineered from the SPA bundle (/assets/index-DC863BID.js, 7.4 MB) - it can change without notice. GET on...	high - machine-readable, no auth, gives every Dam Safety Act regulation PDF link...	200

Source	Kind	What it provides	Relevance	HTTP
NDSA Reg 9 - form, manner and time interval for forwarding instrument readings to the SDSO (s.32(2)) published · Government of India, Official Gazette https://ndsa.gov.in/ndsa-backend/web/sites/default/files/...	standard	The gazette notification of 24/04/2025 prescribing the form, manner and interval in which a dam owner must forward the ANALYSIS of instrument readings to the State Dam Safety Organisation. caveat: SCANNED IMAGE - pypdf extracts 1 character from 2 pages. Needs OCR or manual reading to get the actual prescribed interval and form. Do...	high - the single regulation that turns AquaSync's data pipeline into a statutory...	200
NDSA Reg 8 - minimum instrumentation in specified dams (s.32(1)) published · Government of India, Official Gazette https://ndsa.gov.in/ndsa-backend/web/sites/default/files/...	standard	Gazette notification of 24/04/2024 prescribing the minimum number of sets of instrumentation at each specified dam and the manner of installation. caveat: Scanned image, 1 character extractable from 2 pages. OCR required.	high - defines the legally required sensor floor that AquaSync's telemetry layer can be...	200
NDSA Reg 16 - data requirements of hydro-meteorological stations near specified dams (s.33(1)) published · Government of India, Official Gazette https://ndsa.gov.in/ndsa-backend/web/sites/default/files/...	standard	Gazette notification of 20/05/2024 prescribing what hydro-meteorological data a dam owner must collect at stations in the vicinity of a specified dam. caveat: Scanned image, ZERO extractable characters. Smallest of the four (1 page) so easiest to OCR first.	high - this is the legal definition of the rainfall/met inputs AquaSync's SCS-CN module...	200
The Dam Safety Act, 2021 (Act 41 of 2021) - full text in force 3 · Government of India statute, public domain https://wrd.maharashtra.gov.in/Site/Upload/Pdf/DSA_2021_a...	standard	Complete statute. Sections that matter here: s.18 log books/database; s.30 dam safety unit; s.31 inspections; s.32 instrumentation and forwarding of reading analysis to the SDSO; s.33... caveat: This is a STATE MIRROR, not indiagate.nic.in. The canonical India Code URLs that search engines return -...	high - the primary legal instrument; cite section numbers, not summaries	200
PIB - 'Dam Rehabilitation: Strengthening Infrastructure through Policy and Technology' (15 May 2026) 15 May 202 · Press Information Bureau, Government of India https://static.pib.gov.in/WriteReadData/specificdocs/docu...	doc	8 pp, current official numbers: India 3rd worldwide with 6,628 specified dams (6,545 operational, 83 under construction), ~330 BCM gross storage; 26% (1,681) over 50 years old, 291 over 100; 98.5%... caveat: A PIB backgrounder, not a regulation - use it for framing and figures, cite the Act and the gazette regulations for anything legal....	high - the most current, quotable, single-source framing of the regulatory landscape...	200
NDSA Rapid Risk Screening / DHARMA dashboard API queried li · Government of India public data, no licence stated https://rrsd.ndsa.gov.in/api/dharma/dashboard	api	Open GET returning JSON:... caveat: Aggregate counts only - no per-dam data, no Kerala breakdown, no instrument readings. A sibling endpoint...	medium - proves NDSA does expose public JSON, and quantifies the national inspection...	200
CERT-In Directions under s.70B(6) IT Act 2000, dated 28 April 2022 issued 28 · Government of India, MeitY/CERT-In https://www.cert-in.org.in/PDF/CERT-In_Directions_70B_28....	standard	The binding cyber-security directions: mandatory reporting of specified cyber incidents to CERT-In within SIX HOURS of noticing them; enabling and maintaining logs of all ICT systems for a ROLLING... caveat: Applies to 'service provider, intermediary, data centre, body corporate and Government organisation' - broad enough that...	high - directly sets AquaSync's logging, retention and hosting obligations if it touches...	200

Source	Kind	What it provides	Relevance	HTTP
CEA (Cyber Security in Power Sector) Guidelines, 2021 issued 202 · Central Electricity Authority, Government of India https://www.aegcl.co.in/wp-content/uploads/2022/06/240620...	standard	Central Electricity Authority guidelines binding on all 'responsible entities' in the power supply system - generation utilities, transmission utilities, load dispatch centres, distribution... caveat: This is a STATE UTILITY MIRROR (Assam Electricity Grid Corporation), not cea.nic.in - the CEA site is live (cea.nic.in returns 200) but I...	high - KSEB is a generation utility and Kerala SLDC is a load dispatch centre, so these...	200
CWC - Flood Forecasting / Hydrological Observation live 2026- · Government of India / CWC https://cwc.gov.in/flood-forecasting-hydrological-observa...	doc	The national flood-forecast network: 325 forecast stations covering 197 low-lying areas/cities/towns plus 128 RESERVOIRS, across 20 river basins; 878 hydrological and hydro-meteorological sites... caveat: Narrative page, no station list download, no API. It does NOT confirm whether any Periyar station is a FORECAST station (as opposed to an...	high - establishes the ADVISORY posture (CWC forecasts assist, the dam owner decides)...	200
CWC Advisory Flood Forecast (AFF) - methodology live 2026- · Government of India / CWC https://aff.india-water.gov.in/method.php	doc	AFF methodology: coupled hydrologic (rainfall-runoff) plus hydrodynamic river models; 20 river basin models covering the CWC flood forecasting network; models re-run EVERY THREE HOURS day and night;... caveat: Portal is labelled Beta. No Kerala or Periyar mention on the methodology page - AFF's 20 basin models may not include the Periyar. No...	high - the operational template for how a government-accepted advisory forecast is...	200
OGC SensorThings API Part 1: Sensing, Version 1.1 (OGC 18-088) v1.1 is cu · OGC standard, freely available https://docs.ogc.org/is/18-088/18-088.html	standard	The full Implementation Standard: OData-derived REST model with Thing / Location / Datastream / Sensor / ObservedProperty / Observation / FeatureOfInterest entities, full CRUD,... caveat: NOT adopted by India-WRIS, NHP, CWC or KSDMA - adopting it buys international interoperability, not Indian integration. The data model...	medium - the right standard IF an agency partner demands sensor-observation...	200
OGC API - Environmental Data Retrieval Standard (OGC 19-086r6) published · OGC standard, freely available https://docs.ogc.org/is/19-086r6/19-086r6.html	standard	Lightweight query interface family (position, radius, area, cube, trajectory, corridor, items, locations, instances) for spatio-temporal data - designed exactly for asking 'give me the time series... caveat: Standards conformance is worth nothing on its own if no Indian counterparty consumes EDR - and none currently does. Justify it as...	high - the lowest-effort way to make AquaSync's reservoir-level, discharge and...	200

Source	Kind	What it provides	Relevance	HTTP
pygeoapi 623★ · pushed 202 · MIT https://github.com/geopython/pygeoapi	repo	Python server implementing the OGC API suite (Features, Coverages, Tiles, Records, Processes, EDR) with a plugin provider mechanism; deploys as a Flask/Starlette app over an existing data source. caveat: 0.24.0 requires Python >=3.12 - check AquaSync's interpreter before committing. 40 open issues. You must write an EDR provider plugin for...	high - pure Python, MIT, drops into AquaSync's existing stack, ~1-2 days to a working...	200
FROST-Server (Fraunhofer IOSB) - OGC SensorThings reference implementation 227★ · pushed 202 · LGPL-3.0 https://github.com/FraunhoferIOSB/FROST-Server	repo	The first complete open-source official reference implementation of the OGC SensorThings API, with a plugin system that can extend or replace the data model; ships as Docker images... caveat: Java + PostgreSQL/PostGIS + Docker - a whole second runtime. 86 open issues. The Python client frost-sta-client (LGPL3, 1.1.53, released...	medium - use only if a counterparty mandates SensorThings conformance; otherwise it is a...	200
pymodbus 2762★ · pushed 202 · BSD-3-Clause (GitHub metadata says NOASSERTION; PyPI declares BSD-3-Clause) https://github.com/pymodbus-dev/pymodbus	repo	Full Modbus implementation in Python - client, server and simulator, sync and async APIs, TCP/RTU/ASCII/TLS. caveat: Modbus carries no timestamp, no quality flag and no authentication. Anything you read has the poll time, not the measurement time -...	low for KSEB integration, high as a bench tool - Modbus is not how a dam SCADA in India...	200
opcua-asyncio (asyncua) 1468★ · pushed 202 · LGPL-3.0 https://github.com/FreeOpcUa/opcua-asyncio	repo	Async Python OPC-UA client and server: browsing, subscriptions, historical data access, method calls, security policies and certificate handling. caveat: LGPL-3.0 - fine as an imported library in a service, needs review if AquaSync is ever statically linked or shipped. 244 open issues. The...	high - OPC-UA is the realistic read-only northbound boundary a modern SCADA/EMS will...	200
iec104-python / c104 (Fraunhofer FIT) - IEC 60870-5-104 in Python 101★ · pushed 202 · GPL-3.0 https://github.com/Fraunhofer-FIT-DIEN/iec104-python	repo	Object-oriented Python module wrapping lib60870-C to simulate and speak IEC 60870-5-104 as SCADA master or RTU, with state structures and Python callback handlers. caveat: GPL-3.0 IS A REAL BLOCKER if AquaSync is ever distributed rather than operated as a service - and the underlying mz-automation/lib60870...	medium-high - IEC 101/104 is the protocol family Indian transmission-level SCADA...	200
libiec61850 (MZ Automation) - IEC 61850 including the -7-410 hydro profile 1212★ · pushed 202 · GPL-3.0 (commercial licence available) https://github.com/mz-automation/libiec61850	repo	C implementation of IEC 61850 client/server (MMS, GOOSE, Sampled Values) - the substrate for IEC 61850-7-410, which defines the logical nodes and data objects for hydropower plants (gates, turbines,... caveat: GPL-3.0 with the same distribution problem as lib60870. No maintained Python binding. 243 open issues. Documented IEC 61850 hydro...	low today, strategic later - nothing at Idukki speaks IEC 61850, but -7-410 is the only...	200

Source	Kind	What it provides	Relevance	HTTP
Eclipse Tahu - MQTT Sparkplug B reference implementations 264★ · pushed 202 · EPL-2.0 https://github.com/eclipse-tahu/tahu	repo	Reference client libraries and compatible implementations for the Eclipse Sparkplug specification (currently v3.0), defining MQTT topic namespace, protobuf payload encoding, birth/death certificates... caveat: Tahu itself is JAVA-centric (language=Java). The Python option, matteosox/pysparkplug, is a single-maintainer project - 24 stars, last...	medium-high for AquaSync's OWN edge devices (a stage sensor at Neeleswaram, a Kochi tide...	200

Verification notes

- This domain has NOT been through the adversarial verification pass. URLs carry the researching agent's own reported status only. Treat as provisional.
- dead end: 'OGC HydroDART' DOES NOT EXIST. A targeted search for the exact term returns only WaterML 2.0, HY_Features and GroundwaterML2 - nothing named HydroDART at OGC or anywhere else. This is another fabricated name from the planning material, in the same family as the repeated-ASIN problem. The real OGC water/hydro stack is: SensorThings API v1.1 (OGC 18-088), OGC API - EDR (OGC 19-086r6),...
- dead end: INDIA-WRIS IS UNREACHABLE FROM OUTSIDE INDIA - do not plan an integration you cannot test. <https://indiawris.gov.in/wris/> times out (curl 000 after 21 s, and WebFetch gets ECONNREFUSED on 164.100.85.36:443). Same for the ArcGIS REST tier <https://arc.indiawris.gov.in/server/rest/services?f=json> (000) and <https://indiawris.gov.in/downloads/India-WRIS-HelpManual.pdf> (000). Also dead from...
- dead end: DHARMA AND THE CWC DAM SAFETY PORTAL ARE ALSO GEO-FENCED. <https://damsafety.cwc.gov.in/> and <https://damsafety.cwc.gov.in/dharma/Home1/index.php> return curl 000 and WebFetch ENOTFOUND (damsafety.cwc.gov.in does not even resolve). <https://dharma.cwc.gov.in/> likewise 000. I could not inspect DHARMA directly. What IS reachable: cwc.gov.in itself (200), ndsa.gov.in (200) and...
- dead end: THE PYTHON DNP3 ECOSYSTEM IS ABANDONED - do not spend a day on it. dnp3/opendnp3 is ARCHIVED (335 stars, last push 2022-05-18). ChargePoint/pydnp3: 38 stars, last push 2022-03-28, 22 open issues, and PyPI pydnp3 is still 0.1.0 from 2018-06-07. VOLTTRON/dnp3-python: 20 stars, last push 2024-12-15, 23 open issues. Compounding this, DNP3 is a North American utility convention - Indian...
- dead end: gost/server (Go SensorThings implementation) IS DEAD: 61 stars, last push 2020-07-07, 22 open issues. Six years stale. FROST-Server is the only maintained open SensorThings server.
- dead end: THE CANONICAL INDIA CODE URLs FOR THE DAM SAFETY ACT 2021 RETURN 404, despite being the top search result. Both <https://www.indiacode.nic.in/bitstream/123456789/17002/1/aA2021-41.pdf> and <https://www.indiacode.nic.in/handle/123456789/17002?locale=en> give 404 to curl even with full Chrome headers and a Google referer. Use the Maharashtra WRD mirror...
- dead end: THE C-FLOOD PORTAL HAS NO VERIFIABLE URL. C-FLOOD is real (launched 2 July 2025 by the Union Minister of Jal Shakti; CWC + C-DAC + NRSC; 2-D hydrodynamic; village-level; 48 h ahead; Mahanadi on NSM HPC at C-DAC Pune, Godavari and Tapi from NRSC under NHP) but every plausible portal address I tried failed: <https://cflood.gov.in/> (000) and <https://cflood.cdac.in/> (000). Only the PIB...
- dead end: KSEB'S 2019 JAN-MAY MONTHLY FILES ARE GONE. Package IDs 310-314 on dams.kseb.in 302-redirect to <http://kseboa.in/wp-content/uploads/2020/05/January-2019.xlsx> and kseboa.in does not resolve (curl 000 on both the host root and the file). The usable KSEB archive therefore starts at June 2019 (id 315), not January 2019 as the listing page implies. Do not promise Jan-May 2019.
- dead end: KERALA SLDC'S HISTORICAL DATE FORM DOES NOT WORK HEADLESSLY. sldckerala.com/index.php?id=7 carries a POST form (hidden txtDate=943900200, submit sbtstore=SHOW)

plus a calendar iframe writing a date1 field in a malformed 'YYYY-MM/DD-YY' shape. I tried POSTing txtDate as unix timestamps for 2018/2019/2021/2026 dates and date1 in three formats, to both index.php and index.php?id=7 - every...

- dead end: THE KSEB LIVE DASHBOARD HAS NO USEFUL HISTORY, AND ITS OWN DATE FORMAT IS WRONG. kseb.in/dashboard/view/waterlevel accepts ?date= but ONLY in ISO YYYY-MM-DD; the DD-MM-YYYY value its own renders (26-08-2026) silently returns an empty table, which looks like 'no data for that day' rather than a format error. Verified data exists for 2026-08-01 through 2026-08-26 and...

What is now on disk

Everything below was downloaded into `research/` and verified non-empty. Bulk snapshots are gitignored - they are large and re-fetchable - while the manifest, index and findings are committed so the research is reproducible.

Repository snapshots are extracted from GitHub tarballs rather than cloned. Two of them ship filenames containing colons, which are illegal on Windows and abort a git checkout entirely; tar skips those entries and extracts the remaining files.

Acquired (14)

Resource	Kind	Local path	Size	Why it is here
pastas	repo	<code>research/repos/pastas</code>	164.0 MB	Hydrological time-series analysis. Large; low direct relevance.
google-flood-forecasting	repo	<code>research/repos/google-flood-forecasting</code>	140.1 MB	Google's open-sourced hydrology framework behind Flood Hub.
NOAA-t-route	repo	<code>research/repos/t-route</code>	81.7 MB	Operational Muskingum-Cunge routing used in the US National Water Model.
Sentinel1-Flood-Finder	repo	<code>research/repos/Sentinel1-Flood-Finder</code>	42.9 MB	SAR flood extent mapping. Validates extent, never timing (6-12 day revisit).
opendatakerala-lsg-boundaries	dataset	<code>research/repos/lsg-kerala-data</code>	30.2 MB	Kerala ward/panchayat boundaries as GeoJSON. Needed for downstream exposure mapping.
AI4Water	repo	<code>research/repos/AI4Water</code>	26.3 MB	ML framework for hydrological time series. MIT.
LFPTools	repo	<code>research/repos/LFPTools</code>	24.0 MB	LISFLOOD-FP preprocessing. Last push 2021.
neuralhydrology	repo	<code>research/repos/neuralhydrology</code>	22.9 MB	LSTM rainfall-runoff, the reference implementation. Directly relevant to the forecast-error gap.
Kerala-Dam-Water-Levels	dataset	<code>research/repos/Kerala-Dam-Water-Levels</code>	15.4 MB	The KSEB bulletin feed AquaSync already uses. 18 KSEB dams + irrigation. Verified: covers 2020-08 to 2026-08, NOT 2018.
rtc-tools	repo	<code>research/repos/rtc-tools</code>	6.5 MB	Deltares reservoir optimisation framework. LGPL-3.0, active.
pywr	repo	<code>research/repos/pywr</code>	4.3 MB	Water resource network allocation model. GPL-3.0.
pyflwdir	repo	<code>research/repos/pyflwdir</code>	2.4 MB	Deltares flow-direction / river-network derivation from a DEM. MIT.
dMC-Juniata-hydroDL2	repo	<code>research/repos/dMC-Juniata</code>	159.6 KB	Differentiable Muskingum-Cunge. Small; directly relevant to routing calibration.
idukki-live-bulletin	dataset	<code>research/sources/datasets/kerala_dams_live.json</code>	16.9 KB	Today's bulletin snapshot for all dams.

Refresh everything

```
python scripts/acquire.py # fetch the whole manifest
python scripts/acquire.py --priority high # just the important ones
python scripts/acquire.py --index-only # rebuild the index only
python scripts/build_research_report.py # rebuild this report
```

Method

Thirteen agents in three phases. Six searched one domain each; six adversarial verifiers re-checked those findings; one synthesised.

Phase	Agents	What they did
Research	6	One per domain. Many distinct web searches, primary sources preferred, every URL curl-verified with the real HTTP status recorded, GitHub repos checked against the API for stars, licence and last push.
Verify	6	Adversarial. Re-checked every URL independently, deep-read the most important sources to confirm they provide what was claimed, and rejected anything unconfirmable. Bot-blocked sites (403/503 to automated requests) were confirmed by another route rather than discarded.
Synthesize	1	Cross-domain integration paths, prioritised upgrades tied to specific verified projects, a download manifest, and the evidence against the thesis.

Domains covered

Domain	Confirmed	Rejected
Historical Kerala / India reservoir and flood data, especially pre-2020 (Idukki, Idamalayar, Periyar basin)	30	0
Open-source reservoir operation, optimisation and forecast-informed operations (FIRO)	29	0
Open-source river routing, hydraulics and 2D flood inundation (plus the DEM / hydrography / gauge-data supply chain needed to drive them) for the Periyar basin	30	0
State of the art in dam operation decision support: what is deployed, what fails	30	0
Integrating with existing dam, SCADA and government water systems (KSEB / KSDMA / CWC / NDSA), industrial protocols, open water-data standards, Dam Safety Act 2021 compliance, and cybersecurity constraints	29	0

What this method does not guarantee.

Verification confirms that a URL resolves and that its content matches what was claimed about it. It does not constitute a peer review of the underlying science, and it cannot detect a source that is internally consistent but wrong. Anything load-bearing for a decision should still be read in full before it is relied on.